

Article

Optimal Dispatch of Multi-Coupling Systems Considering Molten Salt Thermal Energy Storage Retrofit and Cost Allocation Under Rapid Load Variations

Niancheng Zhou ¹, Zhenyu Xu ¹, Yuan Chi ^{1,*}, Yao Zou ^{1,2}, Fei Xu ³ and Xuhui Dai ¹

¹ State Key Laboratory of Power Transmission Equipment Technology, School of Electrical Engineering, Chongqing University, Chongqing 400044, China; cee_nczhou@cqu.edu.cn (N.Z.); 202211131141t@stu.cqu.edu.cn (Z.X.); zoyire@foxmail.com (Y.Z.); 20183423@cqu.edu.cn (X.D.)

² College of Physics and Electronic Engineering, Chongqing Normal University, Chongqing 400044, China

³ State Key Laboratory of Control and Simulation of Power System and Generation Equipment, Tsinghua University, Beijing 100084, China; xufei@tsinghua.edu.cn

* Correspondence: chi.yuan@ieee.org; Tel.: +86-178-5215-3552

Abstract: With the rapid growth of renewable energy generation capacity, integrating thermal power units and renewable energy units at the point of common coupling (PCC) as a coupling system (CS) can significantly enhance the operational reliability and economic efficiency of modern power systems. To better reflect the coordinated operational capability of thermal power units with molten salt thermal storage retrofit (TPUMSTSR) in multi-coupling system (MCS) scheduling, this study proposes a multi-stage optimization-based dispatch method for MCSs. First, rapid load variation (RLV) technology and thermal storage retrofitting are combined to establish a thermal power unit operation model incorporating molten salt thermal storage (MSTS) retrofits under RLVs. Based on this, a three-stage economic dispatch optimization method is proposed to maximize the daily comprehensive generation revenue of MCSs while dynamically allocating peak regulation (PR) revenue costs across different time periods to relax and simplify nonlinear objectives and constraints in the optimization process. Finally, a simulation study is conducted on an MCS in northeast China. The results demonstrate that the proposed flexibility retrofit scheme and optimization algorithm enable a more coordinated and economically efficient dispatch strategy.

Keywords: multi-coupling system; rapid load variation; flexibility transformation; molten salt thermal storage; cost allocation

Academic Editor: Francesco Liberato Capiello

Received: 23 February 2025

Revised: 1 April 2025

Accepted: 3 April 2025

Published: 7 April 2025

Citation: Zhou, N.; Xu, Z.; Chi, Y.; Zou, Y.; Xu, F.; Dai, X. Optimal Dispatch of Multi-Coupling Systems Considering Molten Salt Thermal Energy Storage Retrofit and Cost Allocation Under Rapid Load Variations. *Appl. Sci.* **2025**, *15*, 4062. <https://doi.org/10.3390/app15074062>

Copyright: © 2025 by the author. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

With the rapid growth of global renewable energy generation capacity, wind and photovoltaic (PV) power generation pose significant challenges for grid PR and frequency regulation due to their inherent stochasticity [1–3]. In recent years, thermal power generation and renewable energy have been integrated into a CS entity, combining them both physically and in terms of their external outputs. Regions or power sources that have the conditions to form a CS are widely available. For example, in China's Liaoning power grid, the installed capacity of thermal power and renewable energy that can be integrated into a CS at the same grid connection point exceeds 42% [4]. This integrated approach, with the PCC, has become a key method for promoting renewable energy consumption

and participating in deep peak regulation (DPR) [5]. As a novel application scenario in power systems, the CS has become increasingly prevalent in China. Therefore, focusing on the regulated market framework and operational dispatch methodology of China's CSs holds pioneering significance. However, high penetration of renewable energy increasingly requires flexible modifications of thermal power units. Thermal power units face difficulties in simultaneously achieving rapid load changes, a wide range of PR, and lower coal consumption and emissions [6–8]. On the other hand, the involvement of multiple stakeholders in scheduling complicates the modeling of the coordinated operation of the MCS in the large grid. At present, the allocation of ancillary service revenues in China is regulated and primarily managed by national authorities and the power grid in China. The issues related to the distribution of benefits and costs among the various stakeholders within the MCS emerge during the PR process. To further explore the coordinated operation capabilities of MCSs, studying their optimal operational scheduling within the context of flexible modifications to thermal power units is of practical significance.

In promoting renewable energy integration, the flexible transformation of thermal power units forms the foundation for achieving rapid DPR, as well as frequency regulation, in CSs as a whole [9–12]. Accurately describing the physical characteristics and mathematical models of the transformed thermal power units is crucial for the operation of the CS under high renewable energy penetration scenarios. Currently, the flexible transformation schemes for thermal power units primarily focus on three aspects: low-load stable combustion, rapid startup/shutdown, and fast ramping. Rapid startup/shutdown and fast ramping can adjust the coal-fired plant's output in real time [13]. The study [14] mainly relies on coupling with external devices to achieve improvements in minimum load, start-up time, and ramp rate. The study [6] modified the control scheme for steam temperature and fuel demand, and both flexibility and economic performance were improved when the revised control scheme was adopted. However, they increase the mechanical degradation of the unit and raise additional coal consumption and emissions [15]. In contrast, low-load stable combustion, the use of thermal energy storage in coal-fired plants, is widely adopted. According to [16,17], transforming retrofitting with an MSTs system can significantly enhance the depth of load-following capability. The study [18] uses internal computer algorithms to simulate power boilers under fast thermal loads to achieve low-load stable combustion. The studies [17,19] propose achieving internal thermal energy storage by throttling external thermal energy storage of high-/low-pressure heaters, feed water/condensate bypass, and condensate water throttling. However, due to the limitations of storage capacity, it is challenging to achieve RLVs over a wide range of operating conditions. Combining the rapid ramping and molten salt thermal energy storage for thermal power units can address unit wear and tear, as well as economic efficiency, further improving the flexibility transformation performance. Nonetheless, there is limited research on the operational models of TPUMSTSR under RLVs within a CS.

Conventionally, PR ancillary services for thermal power units have been compensated by the government or the grid side [20]. However, with the increasing coordination of MCSs, the existing ancillary service market framework faces challenges in the allocation of benefits among different stakeholders [21]. The study [22] proposes a mechanism for allocating PR costs among renewable energy sources and allocating compensation among PR resources to improve the economic efficiency and PR enthusiasm of independent system operators. The study [23] assumes that the peak shaving compensation of traditional generator sets is linearly correlated with the PR amount to achieve economic scheduling of PR compensation. However, this approach does not fully consider the contribution to PR; hence, it is difficult to encourage active participation of PR resources. Applying previous rules, CSs may adopt scheduling strategies that overlook the internal volatility of wind and PV power generation to maximize their benefits. These strategies would

demand that thermal power units perform continuous PR without considering varying scenarios. Such practices disrupt the healthy development of the market and increase coal consumption, contradicting the low-carbon goals of CS construction. Therefore, there is an urgent need to establish a benefit-sharing model for scheduling that involves MCS entities.

Regarding the algorithm for solving the MCS coordinated optimization problem, the day-ahead scheduling of the system is typically formulated as a model-based optimization problem [24], with the objective of maximizing the total system revenue. Currently, two main architectures are employed to solve this problem: centralized and distributed. Reference [25] proposes a hybrid computational framework using a centralized architecture to address the economic and emission issues of wind and thermal power units. Reference [26] proposed a stochastic day ahead scheduling method of power system considering the flexibility resources of thermal power units, which uses Monte Carlo simulation to deal with the fluctuation of new energy and reduces the prediction error of load and new energy. Reference [27] analyzed the peak shaving process of thermal power units and the impact mechanism of reasonable wind and PV power curtailment on the operation of the power grid. Taking into account the difficulty and overall economy of power grid peak shaving, a scheduling model for maximizing the economic benefits of wind, PV power, and thermal power systems based on the optimal energy curtailment rate was constructed. The centralized architecture utilizes a coordination center for modeling power dispatch, data processing, and computation. However, it struggles with privacy issues during information exchange between the MCS and grid dispatch departments in centralized cooperative operations. Additionally, it faces challenges in efficiently and accurately managing the short-time-scale, large-scale non-convex optimization problems that arise after considering the step-up characteristics of thermal power plants and the distribution of PR auxiliary service revenues. Reference [28] presents a decentralized economic dispatch model for the power system that accounts for wind power, carbon capture thermal power, and carbon emission trading. This model adopts a distributed architecture to protect regional information privacy while achieving economic operation objectives. Reference [29] introduces an optimal scheduling method for a hybrid power generation system, including thermal, wind, and PV power, based on genetic algorithms and two-point estimation to address the non-convex optimization problem in the coordinated operation model. While the distributed architecture reduces the computational burden in nonlinear systems, its communication mechanism's poor scalability limits its application. It is unsuitable for optimization scheduling in large-scale power grids with MCSs across vast geographic areas.

In light of the limitations in the aforementioned studies, the main contributions of this study are summarized as follows:

(1) Firstly, an operational model for thermal power units under RLVs was developed following the retrofitting of MSTs. By analyzing the output variation of thermal power units under such conditions, the beneficial impact of internal retrofitting with MSTs was demonstrated. Constraints for rapid ramping of thermal power unit output were incorporated to quantify the difference in net load caused by MSTs under RLVs and gradual output variation scenarios. Subsequently, a mathematical model was developed to depict the application of MSTs retrofitting in MCS optimization scheduling problems.

(2) Secondly, for the operational scenarios of China's MCSs, a dynamic allocation method for PR revenues was proposed. This approach eliminates the limitation of the conventional method, where PR revenues of thermal power units are provided solely by the grid side and individual users. Instead, revenues are distributed among MCSs. The PR cost allocation (CA) for each CS is dynamically determined by the adjusted generation

volumes based on tiered load rates across different time periods and the total compensation amount.

(3) Finally, regarding algorithmic implementation, to achieve coordinated economic scheduling between the main grid and MCSs under China's regulated electricity market framework, the optimization problem is divided into two sub-problems: the main grid power optimization problem (Stage One) and the economic scheduling optimization problem for CSs (Stages Two and Three). Independent optimization is performed for the main grid and CSs. The economic scheduling optimization for CSs is carried out in a two-stage process to maximize the total revenue. In Stage Two, the total compensation amount for PR is fixed, which simplifies the relaxation techniques and computational methods introduced in Stage Three for addressing the nonlinear CA objectives.

2. The TPUMSTSR Under RLV Model

This chapter proposes a model for TPUMSTSR under RLVs. The model is primarily designed for thermal power units within the CS, requiring flexibility upgrades to enhance the system's operational stability and regulation capability.

2.1. Description of CS Dispatch Method

Each individual CS consists of multiple thermal power units and renewable energy generation facilities of various types. By coupling geographically proximate renewable energy sources and thermal power units through PCC, the system is linked to the main power grid, enabling simultaneous power supply to both local loads and the larger grid. Externally, the CS is treated as an integrated entity, allowing dispatch authorities to manage it as a whole, thereby improving operational efficiency. Internally, the power sources within the CS can be coordinated locally without considering the network topology. The dispatch method is illustrated in Figure 1.

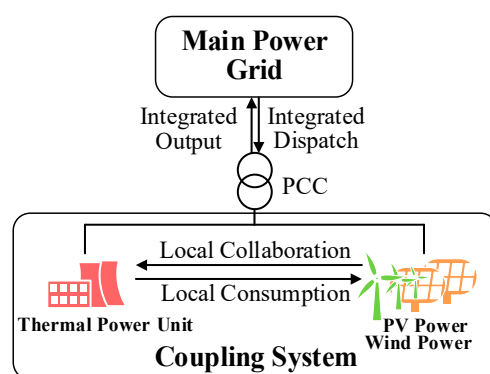


Figure 1. CS dispatch method.

In a similar concept on the source side, the unit combination is mainly determined by the grid dispatching department to calculate the grid flow and determine the coordination strategy for different power sources connected to the grid. Multi-power bundling mainly transmits thermal power and renewable energy power between different regions through high-voltage AC and DC transmission lines but does not emphasize the geographical relationship between power sources. Virtual power plants emphasize the coordination of distributed resources such as energy storage, controllable loads, and electric vehicles on the demand side—rather than the power generation side—to achieve flexible regulation.

Compared with the above categories, the CS coordinates the output of internal power sources based on the dispatch requirements from the main grid's scheduling department, achieving economic operation and enhancing the integration of renewable energy. The CS does not need to consider complex power flows or coordination between power operators in different regions. The large power grid dispatch department does not need to conduct detailed modeling of the power supply within the CS, which simplifies the dispatch difficulty of the dispatch department. At the same time, qualified power sources do not need to make a large amount of additional investment and can directly use existing power grid resources to aggregate into a CS. The thermal power units in the CS can more effectively participate in DPR.

However, since multiple heterogeneous energy sources collectively function as a unified entity within the CS, addressing the intermittency and uncertainty of high-penetration renewable generation necessitates technical modifications to internal thermal power units. Within the current operational framework of CSs, the primary focus is on enhancing the flexibility and response speed of thermal power output regulation.

2.2. Operational Characteristics of Thermal Power Units After RLV Retrofit

To meet the demands of RLVs, thermal power units require equipment, control system, and operational strategy modifications. For example, advanced burner designs and fuel supply systems can be adopted for boiler system retrofitting to improve combustion efficiency and stability during RLVs. According to [13], the ramp rate can be increased to 4% Pe/min when the thermal power high-pressure heater adopts throttling extraction. By optimizing the structure and materials of the boiler and cooling systems, thermal inertia can be reduced, enhancing the boiler's responsiveness to load variations. When the thermal power unit equipment material is upgraded (such as the use of high-temperature alloys), downtime can be reduced by about 15%; however, the initial transformation cost increases by 10–20%. In terms of control system upgrades, advanced control strategies based on artificial intelligence and model predictive control can be introduced to optimize the operational efficiency of the units under RLVs. When the revised control strategy is adopted, the ramp rate can be increased to 7.0% Pe/min, but the maintenance cost can be reduced by 10–15%. These measures enable RLVs, allowing for more timely responses to the variability of stochastic renewable energy generation [30].

Through these technical modifications, thermal power units have acquired rapid ramping capabilities, significantly enhancing their output variation rates and enabling them to quickly respond to the CS load demands. Additionally, these units now possess a wide load operating range, allowing stable operation from low-load to full-load conditions, thereby accommodating the stochastic nature of wind and PV power generation. The RLV capability of thermal power units effectively compensates for the intermittency, improving the operational stability and regulation capacity of the CS.

However, under RLVs, mechanical components such as turbine blades and bearings are subjected to greater thermal and mechanical stresses, leading to accelerated wear and increased lifetime degradation [31]. According to [32], the variable load rate is the main factor influencing equipment service life. Compared with the reference value, increasing the variable load rate by 50% reduces the life of the boiler drum and superheater by 31.9% and 52.9%, respectively, while decreasing the variable load rate by 50% compared with the reference value increases the life of the drum and superheater by 16.3% and 35.8%, respectively. Additionally, existing studies [33] indicate that coal consumption under RLVs undergoes significant changes compared with stable operation due to variations in the boiler's heat storage process and resource utilization during the production of main steam and reheat steam. During RLVs, adjusting the condensate water requires additional work from the pumps, and the operating efficiency of the boiler and turbine decreases,

resulting in increased coal consumption. Therefore, although the output variation rate improves after RLV technical modifications, enabling them to respond quickly to system demands, coal consumption increases by approximately 5% [34], and the thermal stress shock of thick-walled components increases significantly, resulting in a shortening of the service life by approximately 15–20%.

2.3. Consideration of the Operating Principles and Characteristics of TPUMSTSR

Thermal energy storage retrofitting has emerged as a promising solution to address the high coal consumption in thermal power units after RLV modifications. It stores excess heat generated by thermal power units and releases it during peak load periods, enhancing the units' PR capabilities. Furthermore, by optimizing unit operations through thermal energy storage systems, coal consumption and carbon emissions are reduced.

Molten salt can operate stably at high temperatures across various thermal energy storage technologies. This feature makes it suitable for the high-temperature steam systems of thermal power units. Additionally, it offers high energy density and large storage capacity. Economically, molten salt materials are cost-effective, chemically stable, and have a long service life. Therefore, TPUMSTSR provides significant advantages, with thermal efficiency exceeding 90%, and also provides the ability to complete charging and discharging processes within minutes, meeting the demands of rapid PR [35]. MSTS enables thermal power units to have a wider power generation range and has certain functions of typical energy storage systems; however, MSTS differs from energy storage systems. As a part of thermal power, MSTS stores high-temperature hot steam, and when it is necessary to increase the external power output of thermal power, the high-temperature hot steam is applied to thermal power. Therefore, MSTS only stores the thermal energy of the thermal power unit itself. The energy storage system, as an independent large-scale energy storage system, directly provides charging and discharging services, which means that its construction cost is not only relatively more expensive than the MSTS transformation within the thermal power plant, but the system also has lower efficiency. At the same time, there are restrictions on application scenarios and resources, which may conflict with the construction environment of the CS.

The MSTS system consists of thermal storage tanks, heat exchangers, pumps, and pipeline systems. The thermal storage tank is used to store high-temperature molten salt. The heat exchanger transfers heat energy to the working fluid (such as water or steam). The pump and pipeline system ensure the circulation and transport of molten salt. In practice, the thermal storage tank is divided into high- and low-temperature dual-cycle storage tanks. The high-temperature storage tank is positioned between the boiler and the high-pressure cylinder of the steam turbine, while the low-temperature storage tank is placed between the steam turbine's working cylinder and the boiler. Heat charging occurs when the thermal power unit operates at low load or during surplus power generation periods. The high- and low-temperature storage tanks absorb excess thermal power from the boiler and the working cylinder of the steam turbine, respectively, and transfer it to the molten salt via heat exchangers, storing it within the tanks. A portion of the steam is diverted from the high-pressure turbine bypass or the low-pressure extraction port. The low-temperature thermal storage tank captures the underutilized waste heat in the low-temperature section (approximately 200–240 °C) of the turbine cylinder by integrating an auxiliary heat exchanger within the turbine casing. The bypassed steam reduces the flow into the low-pressure turbine, thereby decreasing power generation. Ultimately, this process maximizes the recovery and utilization of thermal energy throughout the entire system. This process reduces the minimum output of the thermal power unit. The heat discharging process takes place during peak load periods or when renewable energy output is insufficient. The stored thermal energy in the tanks is released, with the high-

temperature storage tank increasing the steam turbine's inlet steam flow, while the low-temperature storage tank heats the water, forming high-temperature feedwater that enters the boiler, thereby enhancing the output of the thermal power unit. Ultimately, this system enables changes in the unit's external power output P_t^{TPU} without adjusting the thermal power unit combustion power P_t^{BURN} . The energy flow diagram of the modified thermal power unit is shown in Figure 2.

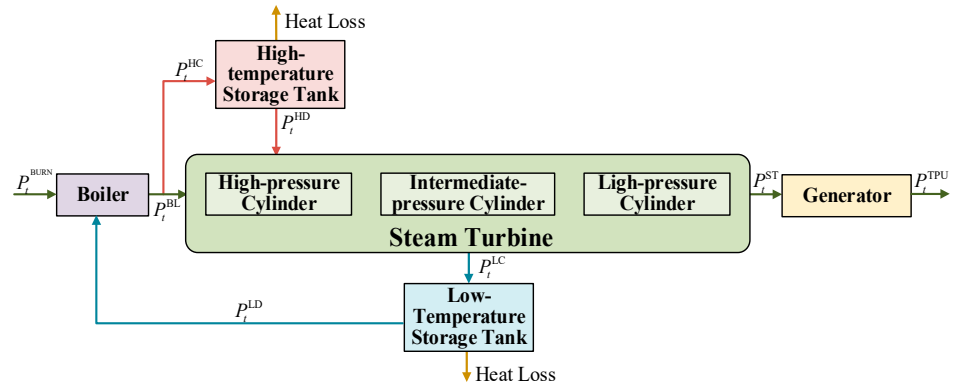


Figure 2. Schematic diagram of internal energy flow in TPUMSTSR.

In Figure 2, P_t^{HC} , P_t^{HD} , P_t^{LC} and P_t^{LD} represent the stored and released thermal power of the high-temperature and low-temperature thermal storage systems; P_t^{BL} represents thermal power input to the steam turbine; and P_t^{ST} represents thermal power output from the steam turbine to the generator.

However, constrained by the thermal storage capacity, charging/discharging time, and efficiency of the thermal storage system, MSTs has limited potential to enhance real-time renewable energy integration and DPR. Its thermal storage capacity is limited by the volume of the storage tanks and the thermal properties of the molten salt. In scenarios with significant fluctuations in renewable energy output or high load demands, the support capacity of thermal storage is limited, making it difficult to fully replace thermal power units retrofitted for RLVs. Additionally, although MSTs systems respond relatively quickly, their charging and discharging times are still constrained by heat exchange efficiency and pipeline transport capacity. Thermal inertia during the charging and discharging processes may cause delays in energy conversion. In scenarios requiring rapid responses (such as sudden changes in wind or PV power output), the thermal storage system may not be able to meet the CS's demands in a timely manner.

2.4. Operational Model of TPUMSTSR Under RLVs

Thermal power units face the dual challenges of reducing carbon emissions and enhancing flexibility. While individual retrofitting technologies (such as MSTs or RLV retrofitting) can partially address these issues, they still have limitations. Integrating MSTs retrofitting with RLV retrofitting can leverage the advantages of both, further improving the PR capability, operational flexibility, and economic efficiency of thermal power units. However, this combined retrofitting approach imposes higher requirements on the operational control of thermal power units, necessitating the establishment of precise mathematical models and consideration of multiple constraints.

To further reduce coal consumption and carbon emissions, the MSTs system is first utilized to regulate the unit's output when thermal power units need to adjust their output to meet dispatch requirements. The operational model of the thermal power unit under this scenario can be described by the following mathematical expression [35,36]:

$$\begin{cases} S_t^H = (1 - \eta^H) S_{t-1}^H + (\eta^{HC} P_t^{HC} - \frac{P_t^{HD}}{\eta^{HD}}) \Delta t \\ S_t^L = (1 - \eta^L) S_{t-1}^L + (\eta^{LC} P_t^{LC} - \frac{P_t^{LD}}{\eta^{LD}}) \Delta t \\ P_t^{BURN} + P_t^{HD} - P_t^{HC} + P_t^{LD} - P_t^{LC} = P_t^{ST} \\ P_t^{BURN} = \frac{\eta^{coal} M_t^{coal} Q_t^{coal}}{\Delta t} \\ P_t^{TPU} = \eta^{ST} P_t^{ST} \\ SOC^{\min} \leq SOC(t) \leq SOC^{\max} \end{cases} \quad (1)$$

where S_t^H and S_t^L represent the stored heat in the high-temperature and low-temperature thermal storage systems at time t ; η^H and η^L denote the heat loss rates of the high-temperature and low-temperature thermal storage systems, respectively; η^{HC} , η^{HD} , η^{LC} , and η^{LD} represent the thermal efficiencies of the high-temperature and low-temperature systems; η^{coal} is the thermal efficiency of coal combustion; M^{coal} is the mass of coal consumed at time t ; Q^{coal} is the calorific value of pulverized coal combustion; η^{ST} represents the thermoelectric conversion efficiency; and SOC (state of charge) represents the remaining power of the energy storage system, described as residual heat in the MSTs system, and is specifically described in the constraint conditions of (30).

To enhance the PR capability and operational flexibility of thermal power units, RLV adjustment is initiated when the capacity or charging/discharging efficiency of the MSTs system cannot meet the load demand. The primary goal in this scenario is to meet the real-time dispatch requirements of the grid. During the operation of the thermal power unit, the following ramp rate constraint under RLVs must be satisfied [3]:

$$\begin{cases} \forall P_{t-1}^{BURN} \in P_a, P_t^{\min, BURN} \in P_b, P_t^{\max, BURN} \in P_c \\ \begin{cases} P_t^{BURN} \leq r_c \Delta T + \frac{r_c}{r_a} P_{t-1}^{BURN} - \sum_{x=c}^{a-1} (\frac{r_c}{r_{x+1}} - \frac{r_c}{r_x}) P_x^{PR} \\ P_t^{BURN} \geq -r_b \Delta T + \frac{r_b}{r_a} P_{t-1}^{BURN} + \sum_{x=a}^{b-1} (\frac{r_b}{r_{x+1}} - \frac{r_b}{r_x}) P_x^{PR} \\ \sum_{x=c}^{a-1} (\frac{r_c}{r_{x+1}} - \frac{r_c}{r_x}) P_x^{PR} = 0, \forall a = c \\ \sum_{x=a}^{b-1} (\frac{r_b}{r_{x+1}} - \frac{r_b}{r_x}) P_x^{PR} = 0, \forall a = b \end{cases} \end{cases} \quad (2)$$

where P_t^{BURN} represents the thermal power unit combustion power; $P_t^{\min, BURN}$ and $P_t^{\max, BURN}$ represent the theoretical minimum and maximum output at time period t ; P_a , P_b , and P_c represent the PR intervals of the thermal power unit; P_x^{PR} denotes the lower limit of the x -th output interval; and r_x is the upper limit of the ramp rate for the x -th output interval.

3. Optimization Dispatch Model for MCSs Considering CA

According to the TPUMSTSR model under RLVs in the previous chapter, the flexibility retrofitted thermal power units can better participate in large-scale and rapid DPR ancillary services within CSs. After incorporating MCSs, compared to traditional grid operation methods, in addition to independent power producers (IPPs) directly connected to the main grid, MCSs will be connected to the main grid through different PCCs, as shown in Figure 3.

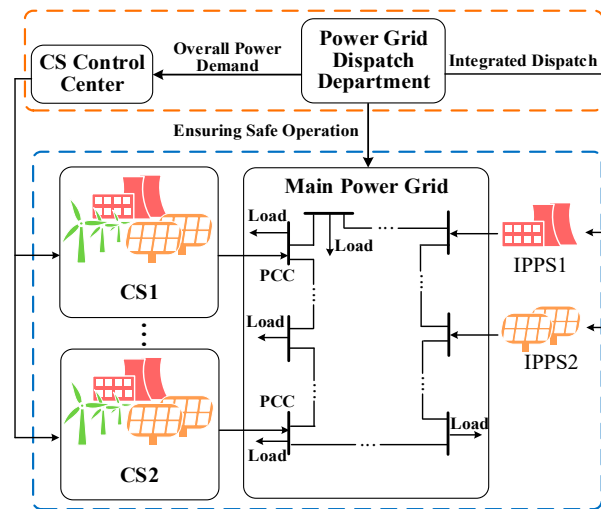


Figure 3. Scheduling method for main grid with MCSs and multi-IPPs.

Under the existing ancillary service market framework, there are no rules for benefit distribution involving MCSs as primary entities. If the traditional compensation mechanism for PR ancillary services, where the state/grid side pays thermal power units, is still applied, it will inevitably lead to CSs adopting dispatch strategies that maximize their own benefits by ignoring the internal volatility of wind and PV power and requiring their thermal power units to continuously provide PR services regardless of the scenario. This behavior not only disrupts the healthy development of the market but also increases coal consumption in thermal power units, undermining the low-carbon goals of CS construction. To prevent uneven benefit distribution among multiple stakeholders, this chapter establishes an economic operation optimization model for MCSs that considers CA.

3.1. CA Model Establishment

Since the primary purpose of PR is to mitigate power fluctuations from wind and PV units within CSs and to incentivize conventional thermal power units to undergo flexibility retrofiting, this chapter establishes a CA calculation method with CSs as the primary stakeholders. The PR revenue obtained by thermal power units is shared by IPPs connected to the same main grid, renewable energy units within the CS, and thermal power units not participating in PR [37]. This approach ensures a fairer and more precisely quantifiable distribution of overall benefits.

The CAs are calculated based on three factors: the total PR compensation revenue of MCSs, the adjusted output of units not participating in PR, and the total adjusted output of all units not participating in PR. These factors are highly nonlinear and involve numerous integer variables, as they are associated with the decision variable “output of individual units.” This significantly increases the complexity of the solution, making it difficult to solve directly using a distributed architecture.

On the other hand, since CS operators and grid dispatch departments belong to different stakeholders, the dispatch department needs detailed models and data from each power source within different operators to schedule each power station in the grid. Traditional centralized optimization methods involve sensitive information such as installed capacity, technical characteristics, and operational status, which may lead to privacy concerns.

To address these issues, this chapter decouples the large-scale mixed-integer nonlinear coordinated operation model of MCSs into two sub-models: the optimal power flow

model of the main grid and the internal operation model of MCSs. The solution is divided into three stages, as illustrated in Figure 4.

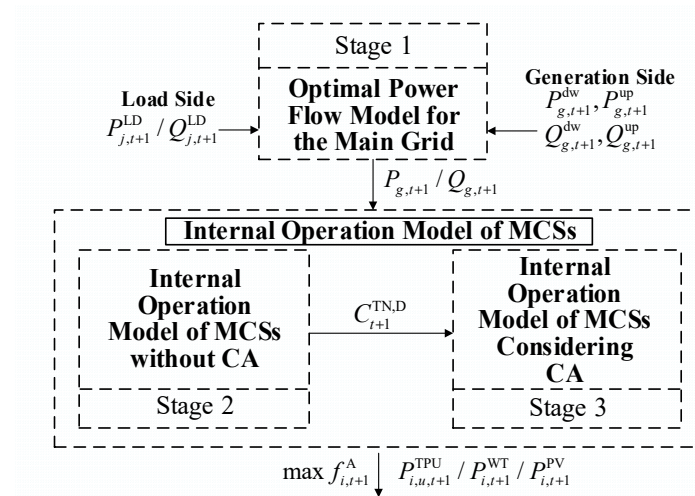


Figure 4. Schematic diagram of the three-stage relationship considering CA.

In Stage One (Main Grid Optimal Power Flow Model), the inputs include the active/reactive power requirements $P_{j,t+1}^{LD}$ and $Q_{j,t+1}^{LD}$ on the load side for the next time period, as well as the feasible power regions $P_{g,t+1}^{up}$, $P_{g,t+1}^{dw}$, $Q_{g,t+1}^{up}$, and $Q_{g,t+1}^{dw}$ of the CSs and IPPs for the next time period. The output of Stage One comprises the active/reactive power that the CSs and IPPs must provide at the next time period, which is then used as input for the internal operation model of the MCS in Stages Two and Three.

By establishing the mathematical expression for CA, it is observed that the CA for thermal power and renewable energy units not participating in PR is a highly nonlinear objective function. Taking the CA for thermal power as an example, the CA for units $f_{i,u,t}^{TP,S}$ not participating in PR is calculated as follows [3]:

$$f_{i,u,t}^{US} = C_t^{TN,D} \frac{P_{i,u,t}^{TPU,C}}{\sum_{g \in G} P_{g,t}^C} \quad (3)$$

where g represents the index of power sources in the main grid; G represents the set of IPPs and MCSs in the main grid; u represents the index of thermal power units within a single CS; U represents the set of thermal power units within a single CS; i represents the index of CSs; I represents the set of CSs; $C_t^{TN,D}$ represents the total revenue for all power sources providing PR auxiliary services at time t ; $P_{i,u,t}^{TPU,C}$ denotes the corrected output of a single thermal power unit; $P_{g,t}^C$ denotes the corrected power output for IPPs and CSs not participating in PR; and $P_{i,u,t}^{TPU,C}$, $C_t^{TN,D}$ and $P_{g,t}^C$ are determined by the decision variables $P_{i,u,t}^{TPU}$, $P_{i,t}^{PV}$, and $P_{i,t}^{WT}$.

In traditional optimization methods, it is challenging to effectively handle the nonlinear objective function involving cross-calculation of fractional and multiplicative variables determined by decision variables. Additionally, due to the fact that when solving for the maximum revenue of a single CS within an MCS, both $C_t^{TN,D}$ and $P_{g,t}^C$ are affected by the output of other units connected to the main grid, traditional optimization methods face significant computational challenges in coordinated optimization. Therefore, this study proposes dividing the overall internal operation model of MCSs into the following two stages.

In Stage Two (Internal Operation Model of the MCSs without CA), the initial total peak-shaving service revenue $C_t^{\text{TN,D}}$ of the MCS for the next time period is determined. This value is then fixed and passed on to Stage Three.

In Stage Three (Internal Operation Model of the MCSs considering CA), the fractional variables are relaxed via second-order cone relaxation. Ultimately, the maximum revenue $\max f_{i,t}^A$ for each coupling system at the next time period is obtained, along with the optimal day-ahead dispatch plans $P_{i,u,t+1}^{\text{TPU}}$, $P_{i,t+1}^{\text{WT}}$, and $P_{i,t+1}^{\text{PV}}$ for each power source in the CSs. CSs must strictly adhere to the dispatch instructions from the main grid, ensuring no over- or under-generation.

3.2. Optimal Power Flow Model for the Main Grid

In the multi-stage optimization operation model of MCSs, Stage One involves decoupling the operation of the main grid and the CSs. The main grid is operated by the dispatch department, where certain independently connected thermal power units are directly controlled by the dispatch department. Meanwhile, MCSs connected to different nodes of the main grid are operated by independent operators.

To achieve hierarchical operation between the main grid and CSs, in Stage One: Optimal Power Flow Model for the Main Grid, the dispatch department determines and sends active and reactive power scheduling requirements over a short time scale to each power source or CS based on the optimal power flow program. Each power source and CS must generate power according to the dispatch requirements at each time period and provide feasible output boundaries for the next time period to facilitate future calculations.

Additionally, the dispatch department, IPPs, and CSs exchange information on DPR auxiliary services, including overall revenue and CA; specifically, IPPs and CSs provide their total PR revenue and total adjusted output to the dispatch department. The total PR revenue generated across the entire grid needs to be allocated among power sources that did not participate in PR, following predefined rules. The dispatch department aggregates the PR revenue data uploaded by IPPs and CSs and then redistributes the total auxiliary service revenue and total adjusted generation across the entire grid to IPPs and CSs for CA and economic optimization of their dispatch plans.

As a result, internal power sources within each CS are coordinated by a central controller, which adjusts internal power output according to the overall dispatch requirements from the dispatch department. To enhance the flexibility of thermal power units and accommodate more renewable energy, internal power sources only need to share essential generation information with the central controller. This minimizes information exchange between the main grid and CSs while achieving the independent and privacy-preserving operation of CSs connected to different nodes of the main grid.

3.2.1. Objective Function

In Stage One, the Optimal Power Flow Model of the Main Grid, the optimization objective is to minimize the power generation $P_{g,t}$ of each IPP and MCS at each time period globally.

$$\min f_t^{\text{OPF}} = \sum_{g \in G} P_{g,t} \quad (4)$$

3.2.2. Constraints

In the main grid, the power injected at any node must satisfy the power balance equation. Additionally, during normal operation, the node voltage deviations and branch line

capacities must meet the operational safety requirements; for specific constraint formulations, refer to [38].

Since the main grid directly dispatches IPPs and CSs, the feasible power region of IPPs and CSs must be constrained in the optimal power flow model. The upper and lower power limits are calculated by the IPPs and CSs, and the results are provided to the main grid.

$$P_{g,t}^{dw} \leq P_{g,t} \leq P_{g,t}^{up}, \forall g \in G \quad (5)$$

$$Q_{g,t}^{dw} \leq Q_{g,t} \leq Q_{g,t}^{up}, \forall g \in G \quad (6)$$

$$\begin{cases} P_{g,t}^{dw} = P_{g,t-1} - R_g \Delta t \\ P_{g,t}^{up} = P_{g,t-1} + R_g \Delta t \end{cases}, \forall g \notin I \quad (7)$$

$$\begin{cases} P_{g,t}^{dw} = \sum_{u \in U_i} \bar{a}_{i,u,t}^{TPU} + \bar{a}_{i,t}^{-WT} + \bar{a}_{i,t}^{-PV} \\ P_{g,t}^{up} = \sum_{u \in U_i} \bar{a}_{i,u,t}^{-TPU} + \bar{a}_{i,t}^{-WT} + \bar{a}_{i,t}^{-PV} \end{cases}, \forall (g=i) \in I \quad (8)$$

$$Q_{g,t}^{dw}, Q_{g,t}^{up} = \pm \sqrt{S_g^2 - P_{g,t}^{up}}, \forall g \notin I \quad (9)$$

$$\begin{aligned} Q_{g,t}^{dw}, Q_{g,t}^{up} = & \sum_{u \in U_i} \bar{Q}_{i,u}^{TPU\pm} (\bar{S}_{i,u}^{TPU}, \bar{a}_{i,u,t}^{-TPU}) \\ & + \bar{Q}_i^{PV\pm} (\bar{S}_i^{PV}, \bar{a}_{i,t}^{-PV}) \\ & + \bar{Q}_i^{WT\pm} (\bar{S}_i^{WT}, \bar{a}_{i,t}^{-WT}), \forall (g=i) \in I \end{aligned} \quad (10)$$

where R_g represents the ramp rate of power source; S_g represents the capacity of power source; $\bar{S}_{i,u}^{TPU}$ represents the capacity of a single thermal power unit; $P_{g,t}^{up}$, $P_{g,t}^{dw}$, $Q_{g,t}^{up}$, and $Q_{g,t}^{dw}$ represent the upper and lower boundaries of the feasible active and reactive power regions at time; $\bar{a}_{i,u,t}^{TPU}$, $\bar{a}_{i,u,t}^{WT}$, and $\bar{a}_{i,u,t}^{PV}$ represent the upper bounds of the feasible active power regions for thermal power, wind power, and PV power, respectively; and $\bar{a}_{i,u,t}^{TPU}$ represents the lower bound of the feasible active power region for thermal power.

3.3. Internal Operation Model of MCSs

After obtaining the short-time-scale minimum power dispatch requirements through the optimal power flow (OPF) model of the main grid, each IPP and CS must generate power in accordance with the dispatch requirements at each time period. Since these entities belong to different stakeholders, this section focuses on the internal operation of a single CS.

3.3.1. Objective Function

For a single CS i , the objective function of internal optimized operation is to maximize the total revenue at each time period, denoted as $f_{i,t}^A$

$$\max f_{i,t}^A = \Delta T \left(\sum_{u \in U_i} f_{i,u,t}^{TPU} + f_{i,t}^{RE} - C^{GU} |P_{i,t}^{UN}| \right) \quad (11)$$

where $P_{i,t}^{\text{UN}}$ represents the difference between the actual total power generation of the CS and the dispatch requirement, and C^{GU} is the penalty factor. The comprehensive power generation revenue of thermal power $f_{i,u,t}^{\text{TPU}}$ is given by the following equation:

$$f_{i,u,t}^{\text{TPU}} = f_{i,u,t}^{\text{UB}} + f_{i,u,t}^{\text{UD}} - f_{i,u,t}^{\text{UO}} - f_{i,u,t}^{\text{UE}} - f_{i,u,t}^{\text{US}} - f_{i,e,t}^{\text{HS}} \quad (12)$$

where $f_{i,u,t}^{\text{UB}}$ represents the basic electricity sales revenue of the thermal power unit; $f_{i,u,t}^{\text{UD}}$ represents the compensation revenue from participating in DPR auxiliary services; $f_{i,u,t}^{\text{UO}}$ represents the operational cost of the thermal power unit; $f_{i,u,t}^{\text{UE}}$ represents the environmental cost due to pollutant emissions; $f_{i,u,t}^{\text{US}}$ represents the CA for PR auxiliary services; $f_{i,u,t}^{\text{HS}}$ represents the comprehensive operating cost of the MSTS system of the thermal power unit, where e denotes the index of TPUMSTSR in the single CS and E represents the set of such units.

The total power generation revenue from renewable energy $f_{i,u,t}^{\text{RE}}$ is given as follows:

$$\begin{aligned} f_{i,t}^{\text{RE}} = & C^{\text{WT}} P_{i,t}^{\text{WT}} + C^{\text{PV}} P_{i,t}^{\text{PV}} \\ & - C^{\text{RE,D}} (P_{i,t}^{\text{WT}*} - P_{i,t}^{\text{WT}} + P_{i,t}^{\text{PV}*} - P_{i,t}^{\text{PV}}) \\ & - f_{i,t}^{\text{WTS}} - f_{i,t}^{\text{PVS}} \end{aligned} \quad (13)$$

where C^{WT} and C^{PV} denote the unit revenue of wind and PV power generation, respectively; $C^{\text{RE,D}}$ denotes the curtailment cost for wind and PV power; $P_{i,t}^{\text{WT}*}$ and $P_{i,t}^{\text{PV}*}$ denote the maximum possible generation capacity of wind and PV power at time t , respectively; $P_{i,t}^{\text{WT}}$ and $P_{i,t}^{\text{PV}}$ represent the actual generation outputs of wind and PV power at time t , respectively; and $f_{i,t}^{\text{WTS}}$ and $f_{i,t}^{\text{PVS}}$ denote the CA for PR auxiliary services of wind and PV power, respectively.

(1) Comprehensive Electricity Sales Revenue of Thermal Power Units $f_{i,u,t}^{\text{UB}} + f_{i,u,t}^{\text{UD}}$

In addition to the basic electricity sales revenue, thermal power units can receive PR auxiliary service compensation when their generation output is below the compensation threshold. Therefore, CSs can leverage their local renewable energy consumption advantages to obtain more PR compensation, thereby increasing overall revenue and enhancing competitiveness.

The total electricity sales revenue of a thermal power unit is expressed as the sum of $f_{i,u,t}^{\text{UB}} + f_{i,u,t}^{\text{UD}}$. When the unit's power generation exceeds the compensation threshold, the revenue consists only of the basic revenue $f_{i,u,t}^{\text{UB}}$, which can be expressed as follows:

$$f_{i,u,t}^{\text{UB}} = C_0 P_{i,u,t}^{\text{TPU}} \quad (14)$$

where $P_{i,u,t}^{\text{TPU}}$ represents the electric power generated by thermal power plants for external use and C_0 is the basic unit electricity sale price, which the thermal power unit can receive regardless of its generation state.

When the unit's power generation is below the compensation threshold, in addition to the basic revenue $f_{i,u,t}^{\text{UB}}$, it also includes DPR auxiliary service compensation revenue $f_{i,u,t}^{\text{UD}}$, given by the following equation:

$$f_{i,u,t}^{\text{UD}} = \begin{cases} 0, & \mu_1 < \mu_{i,u,t}^{\text{TPU}} \leq 1 \\ C_1 k_m^{\text{TPU}} (\mu_1 P_{i,u}^{\text{TPU,max}} - P_{i,u,t}^{\text{TPU}}), & \mu_2 < \mu_{i,u,t}^{\text{TPU}} \leq \mu_1 \\ C_2 k_m^{\text{TPU}} (\mu_1 P_{i,u}^{\text{TPU,max}} - P_{i,u,t}^{\text{TPU}}), & 0 < \mu_{i,u,t}^{\text{TPU}} \leq \mu_2 \end{cases} \quad (15)$$

$$\mu_{i,u,t}^{\text{TPU}} = \frac{P_{i,u,t}^{\text{TPU}}}{P_{i,u}^{\text{TPU,max}}} \quad (16)$$

where $P_{i,u,t}^{\text{TPU,max}}$ is the maximum output of the thermal power unit; C_1 and C_2 are the unit electricity sales prices corresponding to different compensation thresholds, which can only be obtained if the unit's output is below the corresponding threshold; $\mu_{i,u,t}^{\text{TPU}}$ is the load rate of the thermal power unit within the CS; μ_1 and μ_2 are the load rates corresponding to compensation thresholds; and k_m^{TPU} is the seasonal revenue adjustment coefficient.

(2) Operating Costs of Thermal Power Units $f_{i,u,t}^{\text{UO}}$

This cost accounts for two operational states: regular peak regulation (RPR) and DPR. In RPR mode, operating costs only consider the fuel consumption of the unit, whereas in DPR mode, additional losses due to output reduction are also included. The operating cost $f_{i,u,t}^{\text{UO}}$ is calculated as follows:

$$f_{i,u,t}^{\text{UO}} = \begin{cases} C^{\text{coal}} [a_{i,u}^{\text{RPR}} (P_{i,u,t}^{\text{BURN}})^2 + b_{i,u}^{\text{RPR}} P_{i,u,t}^{\text{BURN}} + c_{i,u}^{\text{RPR}}] & \mu_1 < \mu_{i,u,t}^{\text{TPU}} \leq 1 \\ C^{\text{coal}} [a_{i,u}^{\text{RPR}} (P_{i,u,t}^{\text{BURN}})^2 + b_{i,u}^{\text{RPR}} P_{i,u,t}^{\text{BURN}} + c_{i,u}^{\text{RPR}}] & 0 < \mu_{i,u,t}^{\text{TPU}} \leq \mu_1 \\ + a_{i,u}^{\text{DPR}} P_{i,u,t}^{\text{BURN}} + b_{i,u}^{\text{DPR}} & \end{cases} \quad (17)$$

where C^{coal} is the unit price of coal per ton; $a_{i,u}^{\text{RPR}}$, $b_{i,u}^{\text{RPR}}$, and $c_{i,u}^{\text{RPR}}$ are the fitting coefficients for RPR loss; and $a_{i,u}^{\text{DPR}}$ and $b_{i,u}^{\text{DPR}}$ are the fitting coefficients for DPR loss. The operating costs under RPR mode mainly consider coal consumption and purchase costs. Under DPR mode, additional losses incurred due to DPR are also considered [39].

(3) Pollutant Emission Costs of Thermal Power Units $f_{i,u,t}^{\text{UE}}$

Coal-fired thermal power generation results in the emission of various pollutants, necessitating an assessment of pollutant emission costs:

$$f_{i,u,t}^{\text{UE}} = \sum_{h=1}^H c_h^{\text{TE}} \frac{\rho_h^{\text{TE}}}{\tau_h^{\text{TE}}} (a_h^{\text{TE}} P_{i,u,t}^{\text{BURN}} + b_h^{\text{TE}}) \quad (18)$$

where c_h^{TE} , ρ_h^{TE} , and τ_h^{TE} denote the unit treatment cost, unit generation emission factor (kg/MW), and pollutant conversion coefficient, respectively; a_h^{TE} and b_h^{TE} are pollutant fitting coefficients; h represents pollutant types; and H is the total number of pollutants. The considered pollutants include dust, SO_2 , and NO_x .

(4) CA of Thermal Power Units $f_{i,u,t}^{\text{US}}$

The compensation revenue for thermal power units participating in DPR must be shared among external IPPs, non-participating thermal power units within the CS, and renewable energy units. Given the capacity differences among internal power sources in the CS, a fair and reasonable CA method requires determining the corrected output of each internal power source involved in the allocation. Ultimately, the PR compensation revenue is distributed based on the proportion of the corrected output of each power source. Based on the existing ancillary service market rules applicable to generation entities in northeast China [22], the mathematical expression for computing the CA of thermal power units that do not participate in DPR is formulated as follows:

$$f_{i,u,t}^{\text{US}} = \begin{cases} C_t^{\text{TN,D}} \frac{P_{i,u,t}^{\text{TPU,C}}}{\sum_{g \in G} P_{g,t}^{\text{C}}}, \mu_1 < \mu_{i,u,t}^{\text{TPU}} \leq 1 \\ 0, 0 < \mu_{i,u,t}^{\text{TPU}} \leq \mu_1 \end{cases} \quad (19)$$

$C_t^{\text{TN,D}}$ is computed as follows:

$$C_t^{\text{TN,D}} = \sum_{i=1}^I \sum_{u=1}^U f_{i,u,t}^{\text{UD}} \quad (20)$$

$P_{i,u,t}^{\text{TPU,C}}$ is given by the following equation:

$$P_{i,u,t}^{\text{TPU,C}} = \begin{cases} 0, & 0 < \mu_{i,u,t}^{\text{TPU}} \leq \mu_1 \\ z_1^{\text{TPU}} P_{i,u,t}^{\text{TPU}}, & P_{i,u,t}^{\text{TPU}} \in [P_{i,u,t}^{\text{TPU,min}}, A_1 P_{i,u,t}^{\text{TPU,max}}], \mu_{i,u,t}^{\text{TPU}} \geq \mu_1 \\ z_1^{\text{TPU}} A_1 P_{i,u,t}^{\text{TPU,max}} + z_2^{\text{TPU}} (P_{i,u,t}^{\text{TPU}} - A_1 P_{i,u,t}^{\text{TPU,max}}), & P_{i,u,t}^{\text{TPU}} \in [A_1 P_{i,u,t}^{\text{TPU,max}}, A_2 P_{i,u,t}^{\text{TPU,max}}], \mu_{i,u,t}^{\text{TPU}} \geq \mu_1 \\ z_1^{\text{TPU}} A_1 P_{i,u,t}^{\text{TPU,max}} + z_2^{\text{TPU}} (A_2 - A_1) P_{i,u,t}^{\text{TPU,max}} & P_{i,u,t}^{\text{TPU}} \in [A_2 P_{i,u,t}^{\text{TPU,max}}, P_{i,u,t}^{\text{TPU,max}}], \mu_{i,u,t}^{\text{TPU}} \geq \mu \\ + z_3^{\text{TPU}} (P_{i,u,t}^{\text{TPU}} - A_1 P_{i,u,t}^{\text{TPU,max}}), & \end{cases} \quad (21)$$

where A_1 and A_2 represent different allocation tiers; z_1^{TPU} , z_2^{TPU} , and z_3^{TPU} are correction coefficients for thermal power; $P_{i,u,t}^{\text{TPU}}$ is the actual power output; and $P_{i,u,t}^{\text{TPU,max}}$ and $P_{i,u,t}^{\text{TPU,min}}$ are the maximum and minimum power outputs, respectively.

$P_{g,t}^{\text{C}}$ is expressed as follows:

$$\begin{cases} P_{i,t}^{\text{C}} = \sum_{u=1}^U P_{i,u,t}^{\text{TPU,C}} + P_{i,t}^{\text{WT,C}} + P_{i,t}^{\text{PV,C}} \\ P_{g,t}^{\text{C}} = \sum_{i=1}^I P_{i,t}^{\text{C}} + \sum_{d=1}^D P_{d,t}^{\text{C}} \end{cases} \quad (22)$$

where $P_{i,t}^{\text{C}}$ is the corrected output of the CS; d indexes the IPPs connected to the main grid; D is the total number of such sources; and $P_{d,t}^{\text{C}}$ is the corrected output of the d IPP.

(5) Comprehensive Operational Costs of TPUMSTSR Systems $f_{i,e,t}^{\text{HS}}$

The integration of TPUMSTSR systems introduces comprehensive operational costs, including maintenance costs $f_{i,e,t}^{\text{OB}}$ and lifespan degradation costs $f_{i,e,t}^{\text{LB}}$

$$f_{i,e,t}^{\text{HS}} = \sum_{e=1}^E (f_{i,e,t}^{\text{OB}} + f_{i,e,t}^{\text{LB}}) \quad (23)$$

where e represents the index of TPUMSTSR and E denotes the total number of TPUMSTSR.

① Maintenance Costs $f_{i,e,t}^{\text{OB}}$

Maintenance costs are incurred during the charging and discharging processes of TPUMSTSR:

$$f_{i,e,t}^{\text{OB}} = \text{sum} \left(|P_{i,e,t}^{\text{HD}} + P_{i,e,t}^{\text{HC}} + P_{i,e,t}^{\text{LD}} + P_{i,e,t}^{\text{LC}}| \right) C_{i,e}^{\text{OB}} \quad (24)$$

where $C_{i,e}^{\text{OB}}$ is the maintenance price per cycle for the TPUMSTSR system.

② Lifespan Degradation Costs $f_{i,e,t}^{\text{LB}}$

Lifespan degradation costs arise from the wear and tear that occur during the charging and discharging cycles:

$$f_{i,e,t}^{\text{LB}} = \frac{C_{i,e}^{\text{IB}}}{V_{i,e}^{\text{CL}}} \frac{s_{e,t}^{\text{HB}} + s_{e,t}^{\text{LB}}}{2} \quad (25)$$

where $C_{i,e}^{\text{IB}}$ is the initial investment cost, $V_{i,e}^{\text{CL}}$ is the total number of life cycles, and $s_{e,t}^{\text{HB}}/s_{e,t}^{\text{LB}}$ is a binary variable indicating the state change (1 for a change, 0 for no change) of the TPUMSTSR system at time t .

(6) CA for Renewable Energy Systems $f_{i,t}^{\text{WTS}} + f_{i,t}^{\text{PVS}}$

The CA for renewable energy systems, including wind and PV, follows a similar methodology as for TPUs:

$$\begin{cases} f_{i,t}^{\text{PVS}} = C_t^{\text{TN,D}} \cdot \frac{P_{i,t}^{\text{PV,C}}}{\sum_{g \in G} P_{g,t}^{\text{C}}} \\ f_{i,t}^{\text{WTS}} = C_t^{\text{TN,D}} \cdot \frac{P_{i,t}^{\text{WT,C}}}{\sum_{g \in G} P_{g,t}^{\text{C}}} \\ P_{i,t}^{\text{PV,C}} = z_t^{\text{PV,S}} P_{i,t}^{\text{PV}} \\ P_{i,t}^{\text{WT,C}} = z_t^{\text{WT,S}} P_{i,t}^{\text{WT}} \end{cases} \quad (26)$$

where $P_{i,t}^{\text{PV,C}}$ and $P_{i,t}^{\text{WT,C}}$ represent the corrected power outputs of single PV and wind turbine units, respectively.

3.3.2. Constraints

The constraints of the coordinated operation model within the CS can be categorized into the power balance constraint of the CS, the stepped ramp constraint of thermal power units, and others.

(1) Power Balance Constraint of the CS

The sum of power generation from thermal power units and renewable energy within the CS must meet the power demand of the main grid on the CS, expressed as follows:

$$P_{i,t}^{\text{L}} = \sum_{u=1}^U P_{i,u,t}^{\text{TPU}} + P_{i,t}^{\text{WT}} + P_{i,t}^{\text{PV}} + P_{i,t}^{\text{UN}} \quad (27)$$

where $P_{i,t}^{\text{L}}$ is the power demand of the main grid on the CS during the t dispatch period, and $P_{i,t}^{\text{UN}}$ represents the active power imbalance between the total power generation and the dispatch command.

(2) Stepped Ramp Constraint of Thermal Power Units

Thermal power units exhibit different ramp rates under different operating states, known as the stepped ramp characteristic. The feasible power output region between adjacent time periods t and $t + 1$ is correlated with the stepped ramp characteristics, the output range under different states, and the selection of time scales. For the non-convex stepped ramp constraints of thermal power units, refer to Equation (2).

(3) Power Generation Constraints of Thermal Power Units

The power generation constraints of thermal power units include the upper and lower output limits and the flexible reserve constraints, as follows:

① Upper and Lower Output Limits of Thermal Power Units

The power output of thermal power units during each dispatch period must remain within their allowable output range:

$$P_{i,u}^{\text{TPU,min}} \leq P_{i,u,t}^{\text{TPU}} \leq P_{i,u}^{\text{TPU,max}} \quad (28)$$

② Flexible Reserve Constraint of Thermal Power Units

Thermal power units must reserve a certain amount of spinning reserve during each dispatch period to address uncertainties and power imbalances between generation and load demand during the operation of the CS:

$$\begin{cases} P_{i,u,t}^{\text{TPU}} = P_{i,u,t}^{\text{TPU,B}} + P_{i,u,t}^{\text{TPU,F}} \\ -\tau_{i,u}^{\text{TPU}} P_{i,u}^{\text{TPU,max}} \Delta T \leq P_{i,u,t}^{\text{TPU,F}} \leq \tau_{i,u}^{\text{TPU}} P_{i,u}^{\text{TPU,max}} \Delta T \end{cases} \quad (29)$$

where $P_{i,u,t}^{\text{TPU,B}}$ is the predetermined fixed base output of the unit; $P_{i,u,t}^{\text{TPU,F}}$ is the real-time flexible output of the unit to mitigate source and load fluctuations, which can be adjusted within a certain range to track real-time fluctuations; and $\tau_{i,u}^{\text{TPU}}$ is the coefficient of adjustable flexible resources of the unit.

(4) Constraints of the MSTS Systems

The constraints of the molten salt thermal energy storage system include charging/discharging constraints and state-of-charge constraints [35,36], as follows:

① Capacity Constraints of High/Low-Temperature Thermal Storage Systems

$$\begin{cases} S_{\min}^H \leq S_t^H \leq S_{\max}^H \\ S_{\min}^L \leq S_t^L \leq S_{\max}^L \end{cases} \quad (30)$$

where $S_{\min}^H$ and $S_{\max}^H$ represent the minimum and maximum storage capacities of the high-temperature system during period t , respectively; and $S_{\min}^L$ and $S_{\max}^L$ represent the minimum and maximum storage capacities of the low-temperature system during period t , respectively.

② State Constraints of High/Low-Temperature Thermal Storage Systems

$$\begin{cases} \mu_t^{\text{HD}} + \mu_t^{\text{HC}} \leq 1 \\ \mu_t^{\text{LD}} + \mu_t^{\text{LC}} \leq 1 \\ \mu_t^{\text{HD}} \leq \mu_t^{\text{coal}}, \mu_t^{\text{HC}} \leq \mu_t^{\text{coal}} \\ \mu_t^{\text{LD}} \leq \mu_t^{\text{coal}}, \mu_t^{\text{LC}} \leq \mu_t^{\text{coal}} \\ 0 \leq P_t^{\text{HD}} \leq \mu_t^{\text{HD}} \theta^{\text{HD}} P_t^{\text{BURN}} \\ 0 \leq P_t^{\text{HC}} \leq \mu_t^{\text{HC}} P_{\max}^{\text{HC}} \\ 0 \leq P_t^{\text{LD}} \leq \mu_t^{\text{LD}} \theta^{\text{LD}} P_t^{\text{BURN}} \\ 0 \leq P_t^{\text{LC}} \leq \mu_t^{\text{LC}} P_{\max}^{\text{LC}} \end{cases} \quad (31)$$

where μ_t^{HD} , μ_t^{HC} , μ_t^{LD} and μ_t^{LC} represent the operating states of the high-/low-temperature thermal storage systems during period t ; θ^{HD} and θ^{LD} are the maximum heat release coefficients of the high-/low-temperature systems; and $P_{\max}^{\text{HC}}$ and $P_{\max}^{\text{LC}}$ are the maximum heat release powers of the high-/low-temperature systems, respectively.

(5) Power Generation Constraints of Renewable Energy

Renewable energy generation within the CS can reduce net load variability while enhancing economic efficiency of the system by actively reducing part of the power generation. However, the reduction in power output during each period and over a day must not exceed the specified limits. The power generation constraints are as follows:

$$\begin{cases} \mu^{\text{RE}} P_{i,t}^{\text{RE,pre}} \leq P_{i,t}^{\text{RE}} \leq P_{i,t}^{\text{RE,pre}} \\ \mu^{\text{RE,d}} \sum_{t=1}^T P_{i,t}^{\text{RE,pre}} \leq P_{i,t}^{\text{RE}} \leq \sum_{t=1}^T P_{i,t}^{\text{RE,pre}} \end{cases} \quad (32)$$

where μ^{RE} is the minimum utilization rate of the unit during the t dispatch period, used to limit the maximum power reduction and avoid energy waste, including μ^{WT} and μ^{PV} ; $\mu^{\text{RE,d}}$ is the proportion of the unit's reducible output to the predicted output over a day, including $\mu^{\text{WT,d}}$ and $\mu^{\text{PV,d}}$; and $P_{i,t}^{\text{RE,pre}}$ is the predicted power output of the unit during the t dispatch period.

3.4. Solution Method/Algorithm

The solution process for this model is illustrated in Figure 5.

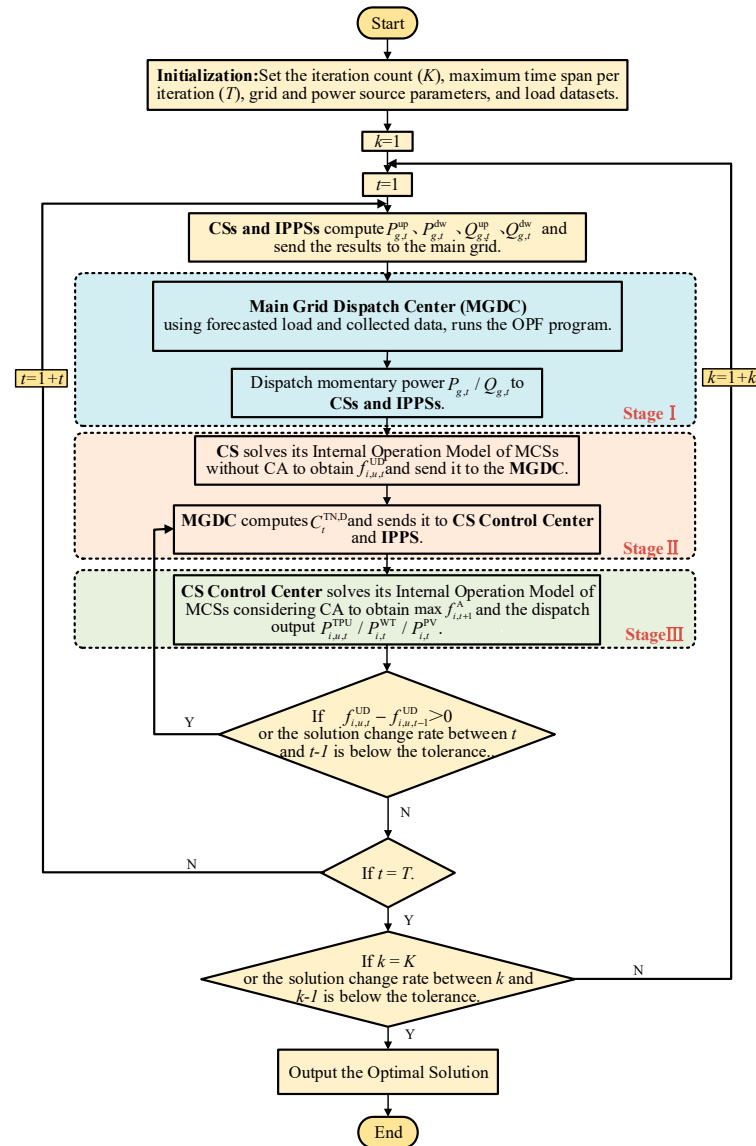


Figure 5. Computation flowchart for economic operation of three-stage MCS considering CA.

(1) Stage One: Solving the Optimal Power Flow Model of the Main Grid

Each grid-connected CS and independently dispatched power source calculates its own upper and lower power output limits for the current time period. Those limits are then transferred to the main grid. After receiving the feasible power region constraints from all participants, the main grid dispatch center solves for the minimum required power generation for each grid-connected CS and IPP based on the predicted load demand. This information is then provided to the corresponding CSs and IPPs.

(2) Stage Two: Solving the Total PR Revenue $C_i^{TN,D}$

Since CA is essentially the distribution of the compensation revenue, in terms of the time sequence, it is possible to first compute the peak-shaving compensation revenue and then calculate the CA for the units that do not participate in peak shaving. Therefore, in Stage Two, the internal operating model of the MCS without CA is solved, and the computed $C_i^{TN,D}$ is fixed. In other words, $C_i^{TN,D}$ can initially be approximated as a constant representing the PR revenue and then incorporated into Stage Three for further optimization.

After solving Stage Two, the nonlinear objective involving cross-calculations of fractional and multiplicative variables in the original CA is partially linearized, significantly reducing the solution difficulty and computation time.

(3) Stage Three: Solving the Economic Dispatch Model of CSs Considering CA

After introducing CA, the PR compensation revenue $C_t^{\text{TN,D}}$ is fixed as a constant in Stage Two. Therefore, the CA calculation only needs to address the fractional nonlinearity of the ratio of the adjusted output of a single unit to the total adjusted output of all units in the system.

To handle the fractional nonlinearity in the objective function, this paper employs second-order cone programming (SOCP) for relaxation. Since the fractional nonlinearity in the CA calculation for thermal power and renewable energy units only differs in the numerator (adjusted output), the following example uses the CA for thermal power units not participating in PR to demonstrate SOCP relaxation:

$$f_{i,u,t}^{\text{US}} = C_t^{\text{TN,D}} \frac{P_{i,u,t}^{\text{TPU,C}}}{\sum_{g \in G} P_{g,t}^{\text{C}}} \quad (33)$$

Let $P_{i,u,t}^{\text{TPU,C}} = A \sum_{g \in G} P_{g,t}^{\text{C}}$, $A^2 = \alpha$, and $(\sum_{g \in G} P_{g,t}^{\text{C}})^2 = \beta$. The original nonlinear CA problem can be described as follows:

$$f_{i,u,t}^{\text{US}} = C_t^{\text{TN,D}} A \quad (34)$$

Additionally, the following second-order cone and variable bound constraints are introduced:

$$s.t. \begin{cases} (P_{i,u,t}^{\text{TPU,C}})^2 \leq \alpha\beta \\ A^2 \leq \alpha \\ (\sum_{g \in G} P_{g,t}^{\text{C}})^2 \leq \beta \\ \sum_{g \in G} P_{g,t}^{\text{C}} \leq \sum_{g \in G} P_{g,t}^{\text{C}} \leq \overline{\sum_{g \in G} P_{g,t}^{\text{C}}} \\ P_{i,u,t}^{\text{TPU,C}} \leq P_{i,u,t}^{\text{TPU,C}} \leq \overline{P_{i,u,t}^{\text{TPU,C}}} \end{cases} \quad (35)$$

After relaxation, the linear CA makes the solution of the optimal dispatch scheme for MCSs convenient. To minimize the error introduced by $C_t^{\text{TN,D}}$ in Stage Two, the PR compensation revenue $f_{i,u,t}^{\text{UD}}$ from Stage Three is summed to obtain the total compensation revenue $C_t^{\text{TN,D,III}}$, which is then compared with the fixed $C_t^{\text{TN,D}}$ from Stage Two. If $C_t^{\text{TN,D,III}} \geq C_t^{\text{TN,D}}$, the constant $C_t^{\text{TN,D}}$ used in Stage Three is updated to $C_t^{\text{TN,D,III}}$, and the Stage Three optimization model is solved again until $C_t^{\text{TN,D,III}} = C_t^{\text{TN,D}}$. This iterative process ensures the accuracy of the multi-stage solution for MCSs and yields the final optimal dispatch scheme.

4. Case Study

4.1. Test System and Parameter Settings

To validate the proposed model, a local grid in northeast China is used as a test system. The empirical data from this validation case exhibit a relatively low renewable energy penetration rate. However, MSTs demonstrates superior economic performance in scenarios with high renewable penetration. The power sources connected to the main grid comprise four CSs and five IPPs that are directly integrated into the grid without flexibility retrofitting. Each CS consists of two coal-fired generating units, one PV station, and one wind power plant. The specific composition of the case study power sources is detailed in Table 1. The scheduling time step (ΔT) is set to 30 min, with a total rolling

scheduling horizon (T_s) of 24 h, resulting in a total of $N_{Ts} = 48$ scheduling periods. Under this short-term scheduling framework, the short-term expected values of renewable energy generation are updated and corrected in real time at each time period, significantly reducing the impact of uncertainty in wind and PV power forecasting within the CS. The OPF voltage operational range is constrained within 0.94 p.u.–1.06 p.u. The DPR parameters are derived from [22]. Table 2 contains other parameters for the CS.

Table 1. Generation settings of the test system.

Power Source	Thermal Power Capacity (MW)	PV Power Station Capacity (MW)	Wind Power Plant Capacity (MW)	
CS1	200+300	200	300	
CS2	300+300	120	180	
CS3	300+600	200	200	
CS4	600+600	400	300	
IPP 1-5 Thermal Power Capacity (MW)				
IPP 1	IPP 2	IPP 3	IPP 4	IPP 5
1040	508	687	580	564

Table 2. Parameters used in the CS.

Parameters	Explanations	Value
A_1, A_2	Different allocation tiers	0.7/0.8
μ_1, μ_2	The load rates corresponding to compensation thresholds	0.5/0.4
$z_1^{TPU}, z_2^{TPU}, z_3^{TPU}$	Correction coefficients for thermal power	1/1.5/2
k_m^{TPU}	The PR revenue correction coefficient	0.5/1
$z_t^{WT,S}$	The wind power output correction coefficient	0.8/1.6
$z_t^{PV,S}$	The PV output correction coefficient	1/2
C_{GU}	The penalty factor	5×10^4 (CNY/MW)
C^{WT}, C^{PV}	The unit revenue of wind and PV power generation	850/740 (CNY/MW)
$C_{i,e}^{OB}$	The maintenance price per cycle for the TPUM-STSR system	10 (CNY/MWh)
μ^{WT}, μ^{PV}	The minimum utilization rate of the unit	0.85/0.80
C_0, C_1, C_2	The unit electricity sales prices corresponding to different compensation thresholds	375/400/1000 (CNY/MW)
C^{WT}, C^{PV}	The unit revenue of wind and PV power generation	850/740 (CNY/MWh)
C^{coal}	The unit price of coal per ton	685 (CNY/t)
C_h^{TE}	The unit treatment cost	1.2 (CNY/kg)
$C^{RE,D}$	The curtailment cost for wind and PV power	10^4 (CNY/MW)
$\tau_{i,u}^{TPU}$	The coefficient of adjustable flexible resources of the unit	0.4
$C_{i,e}^{IB}$	The initial investment cost	4,912,600 (CNY/MWh)
$V_{i,e}^{CL}$	The total number of life cycles	20,000 times

4.2. Economic Analysis of the MSTSR Retrofit Scheme for Thermal Power Units Under RLVs

To compare the proposed TPUMSTSR model under RLVs with the conventional single flexibility retrofitting model, CS 4 from the MCS setup is selected for case study simulation analysis. The case study includes the following three flexibility retrofitting schemes:

Scheme 1: Thermal power undergoes independent RLV retrofitting, with a maximum achievable ramp rate of 1.8 P_N/h for conventional PR and 0.9 P_N/h for DPR.

Scheme 2: Thermal power undergoes independent MSTS retrofitting, with rated capacities of 200 MWh for both high-temperature and low-temperature storage, a heat storage and release efficiency of 98.5%, a thermal-to-electric conversion efficiency of 40%, and maximum heat release coefficients of 0.29 and 0.39 for high-temperature and low-temperature storage, respectively.

Scheme 3: TPUMSTSR under RLVs, with RLVs and thermal storage parameters based on those in Scheme 1 and Scheme 2.

To maximize CS profit, the Gurobi solver is employed to solve the day-ahead optimal scheduling model for a single CS. The various output curves for the 48 time periods from 00:00 to 24:00 are shown in Figures 6–8, while the optimization results for the three retrofitting schemes are summarized in Table 3.

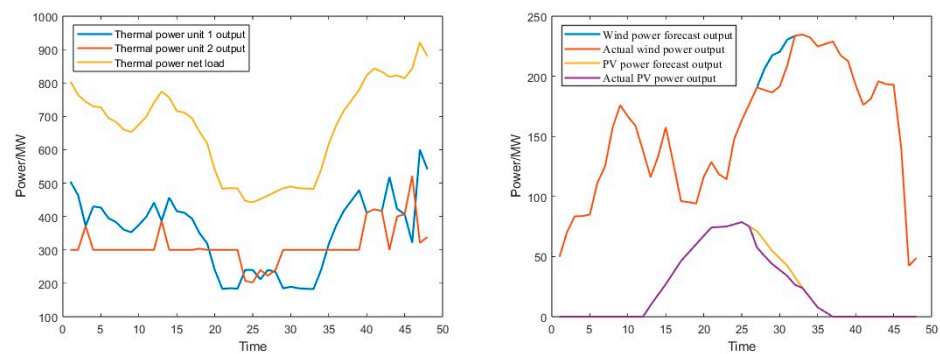


Figure 6. The thermal power output and net load curves, along with the predicted and actual output curves for wind and PV power in Scheme 1.

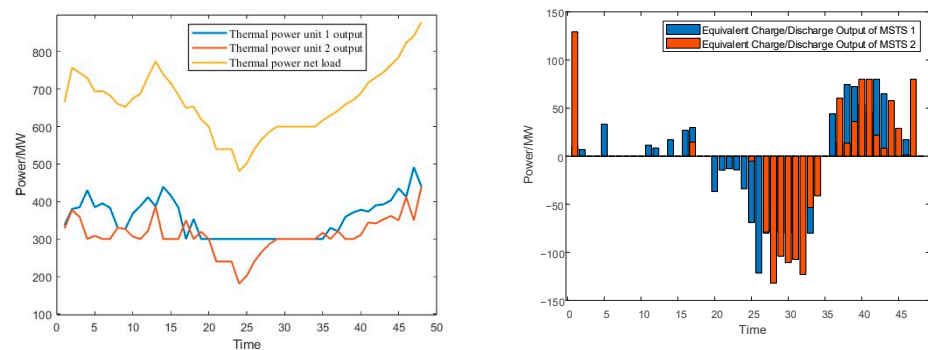


Figure 7. The thermal power output and net load curves, along with the thermal storage system output bar chart in Scheme 2.

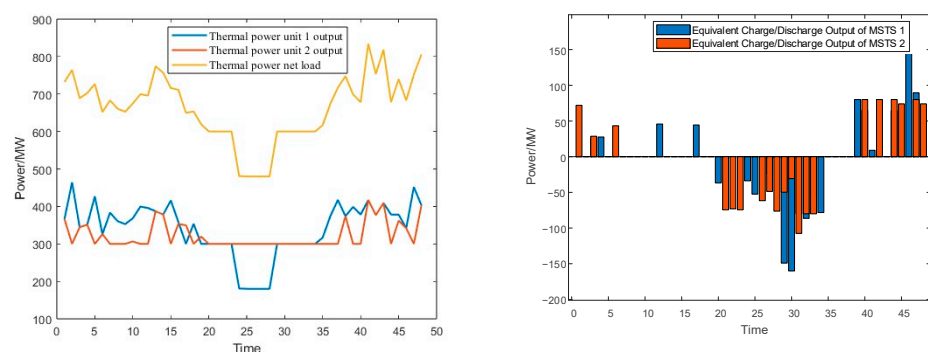


Figure 8. The thermal power output and net load curves, along with the thermal storage system output bar chart in Scheme 3.

Table 3. The optimized revenue and cost results for the CS under the three schemes (CNY).

Category	Operation and Maintenance Costs	Flexibility Reserve Costs	Pollution Costs
Scheme 1	5,626,987.09	41,115.37	1900.02
Scheme 2	5,280,215.66	40,272.32	1901.94
Scheme 3	5,298,865.02	38,006.62	1896.77
Category	Thermal Storage Costs	PR Benefits	Total Benefits
Scheme 1	0	6,492,766.83	4,346,768.08
Scheme 2	105,089.79	6,674,692.46	4,826,702.50
Scheme 3	68,817.64	6,727,512.16	4,899,415.86

The following conclusions can be drawn based on the aforementioned figures and tables:

When thermal power units undergo only RLV retrofitting, their maximum ramping rate between adjacent time periods reaches 9.29 MW/min, significantly improving their real-time ability to accommodate renewable energy. However, the coal consumption and operational maintenance costs under RLVs are relatively high. Additionally, since thermal power units cannot provide sufficient flexibility reserves during DPR, the cost of flexibility reserves increases. To maximize the overall economic benefits of the CS, a small amount of wind and PV curtailment is necessary, as illustrated in Figure 6.

When thermal power units are retrofitted solely with MSTs technology, their maximum ramping rate between adjacent time periods is 3.28 MW/min, resulting in weaker real-time response to renewable energy fluctuations. However, the coal consumption and operational maintenance costs are significantly reduced. Figure 7 shows that the MSTs system charges during DPR valley periods, participating in DPR ancillary services, and discharges during normal or full-load operation. This leads to a significant increase in PR revenue. Additionally, the net load fluctuations of the thermal power unit are substantially reduced, thereby decreasing flexibility reserve costs and significantly increasing total revenue compared to Scheme 1.

When the TPUMSTSR under RLV model is used, the maximum ramping rate is 5.10 MW/min, which is higher than in Scheme 2, thereby enhancing the unit's ability to accommodate renewable energy in real time and further reducing flexibility reserve costs. Figure 8 shows that the overall energy storage output in Scheme 3 is lower, leading to a reduction in energy storage costs. However, due to RLVs, the coal consumption costs are slightly higher than in Scheme 2. At the same time, the RLV technology compensates for the limitations of thermal efficiency and storage capacity in the MSTs system, allowing the unit to participate more frequently in DPR ancillary services. This increases PR revenue, leading to higher total revenue compared to Scheme 2.

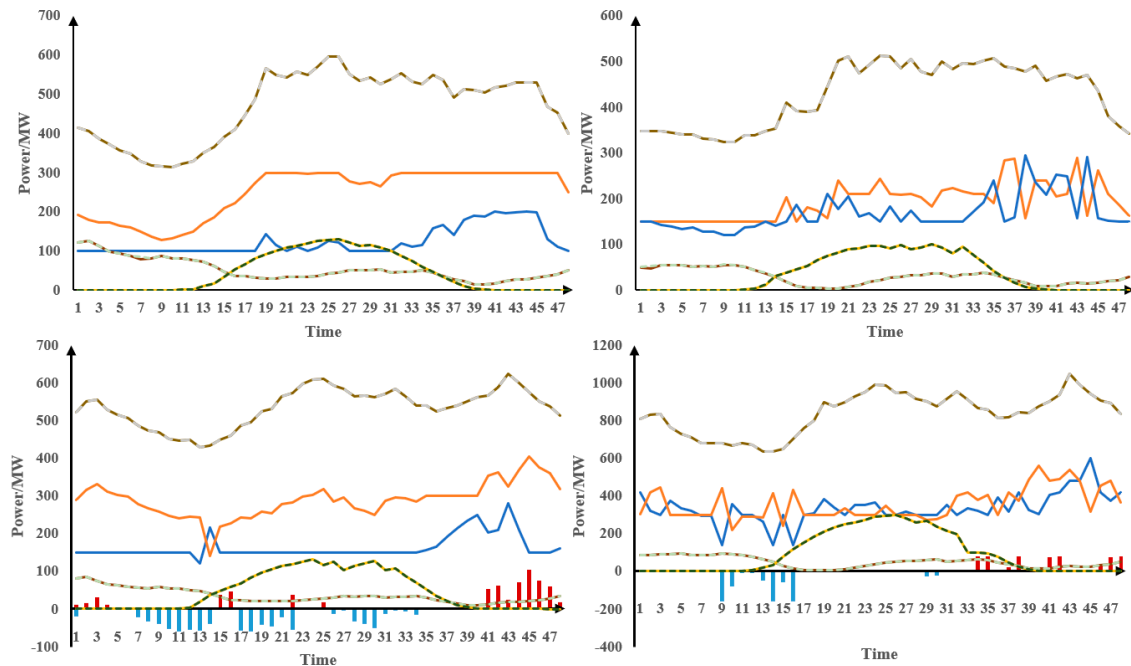
In summary, the proposed TPUMSTSR under RLV model enhances the economic efficiency of the CS, yielding higher overall operational revenue, improving renewable energy accommodation capability, and reducing the peak–valley difference of the net load for thermal power units. In this case study, the total profit improvement of Scheme 3 compared to Scheme 2 is relatively small. This is mainly due to the relatively small fluctuations and peak–valley differences in the current load demand curve, which does not fully reflect the advantages of RLV retrofitting. Additionally, the mechanical wear and tear costs of thermal power units are not included in the day-ahead optimization model for a short time scale. However, the proposed retrofitting scheme is beneficial in reducing mechanical wear and extending the service life of thermal power units.

From a long-term perspective, considering the increasing requirements for real-time power output adjustments in future coal-fired flexibility retrofits, TPUMSTSR under RLVs will demonstrate greater advantages over extended time scales.

4.3. Analysis of the Thermal Energy Storage System in Power Units

In this section, a 600 MW thermal power unit in CSs 3 and 4 is equipped with TPUM-STSR under RLVs, with parameters consistent with Scheme 2 from the previous section. To investigate the effectiveness of multi-stage economic dispatch in an MCS, the Gurobi solver is employed to solve the decoupled and nonlinear relaxed model. To verify the robustness of the method proposed in this study, typical summer and winter days with different load curves and renewable energy output are selected for a case study; specifically, for the summer non-heating period (May to October), the PR revenue correction coefficient $k_m^{\text{TPU}} = 0.5$, the wind power output correction coefficient $z_t^{\text{WT},S} = 0.8$, and the PV output correction coefficient $z_t^{\text{PV},S} = 1$; for the winter heating period (November to April of the following year), these coefficients are set to $k_m^{\text{TPU}} = 1$, $z_t^{\text{WT},S} = 1.6$, and $z_t^{\text{PV},S} = 2$.

The simulation was carried out on a desktop workstation with an AMD Ryzen 7 5700G CPU @ 3.8 GHz and 16 GB RAM. The overall computation time was 588 s (9.8 min) and the total number of iterations was 400,000. The computation time for each MCS was less than 10 min, meeting the real-time dispatch requirements. The detailed power outputs of various generators over two typical days are illustrated in Figure 9 (thermal power unit numbering follows the sequence in Table 2).



(a) Typical summer day

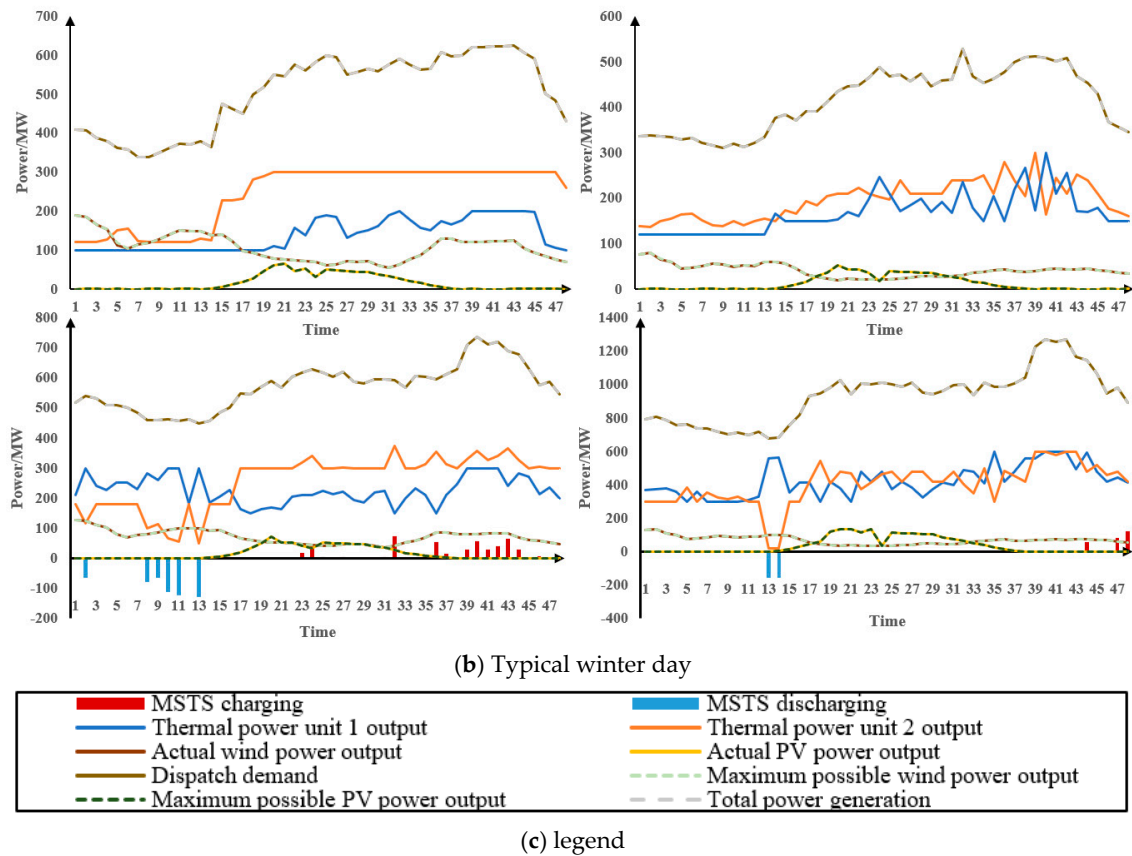


Figure 9. Outputs at each time on two typical days obtained using the three-stage optimization method.

As shown in Figure 9, the upper-level subgraphs represent the output of the internal power sources of coupling systems 1 and 2 from left to right, respectively, and the lower-level subgraphs represent the output of the internal power sources of coupling systems 3 and 4 from left to right, respectively. The power output of each CS aligns with dispatch demands at each time period on two typical days, and the wind/PV generation, follow a similar trend to their maximum output curves. This demonstrates that the proposed three-stage optimization approach effectively maintains dispatch feasibility while preserving privacy among multiple entities, achieving efficient coordination of coal-fired and renewable units. In the typical summer scenario, wind power output within each CS remains relatively low throughout the day, generally staying within 100 MW. PV output starts increasing at 5:00 a.m., reaching its peak (100–300 MW) between 11:00 a.m. and 1:00 p.m. Dispatch demand remains low from 1:00 a.m. to 7:00 a.m., increases from 7:00 a.m. to 10:00 a.m., and stabilizes between 11:00 a.m. and 11:00 p.m. The midday renewable energy output can offset part of the dispatch demand, but thermal power units primarily balance the load throughout the day. Only from 4:00 a.m. to 8:00 a.m. do CSs with flexibility-retrofitted units participate in DPR, leading to lower CA. Simultaneously, the corrected output imposes constraints on DPR, thereby preventing the CS from universally mandating that its internally flexible retrofitted units continuously engage in DPR solely as a profit-seeking mechanism and fostering a healthy competitive market environment. Owing to the relatively low value of k_m^{TU} during summer and the modest DPR revenue from thermal power units, DPR is infrequently employed to reduce operational losses. Meanwhile, in CSs 3 and 4, the higher charge/discharge frequency of the MSTS system expands the electrical output range of the thermal power units, thereby extending the duration of their

participation in DPR. Consequently, the DPR revenue for each CS is enhanced, and the corresponding CA is also increased.

In the typical winter scenario, wind power output in each CS remains high during most periods, with a slight decrease at midday. During this time, PV generation, which experiences a relative increase at midday, compensates for the dip in wind output, resulting in overall renewable energy output that is comparable to that observed in summer. Dispatch demand exhibits greater fluctuations at 20:00 and 22:00 in winter, leading to higher net load requirements for thermal power units. Consequently, the MSTs system is discharging at this time, which increases thermal power output. However, between 00:00 and 08:00, the net load on thermal power units is relatively low due to elevated wind power output, providing more opportunities for participation in DPR. During this period, the MSTs system engages in charging, further enhancing the DPR revenue for thermal power units. Moreover, due to the influence of varying time periods and the stepped load rate adjustments on corrected output, the value of k_m^{TPU} is higher in winter. As a result, the revenue obtainable from DPR participation by thermal power units increases, ultimately incentivizing their involvement in PR. Consequently, the associated CA is relatively higher.

Table 4 presents the total PR revenue and CA for each CS on a typical winter day, thereby quantifying the impact of CA on benefit distribution. Specifically, CSs 3 and 4 achieve higher PR revenue and exhibit a lower ratio of CA to revenue, whereas CSs 1 and 2 show the opposite trend. This disparity is attributed to the fact that CSs 3 and 4 are equipped with TPUMSTSR that possesses RLV capability and a wider adjustable output range. Observations indicate that CS 3 maintains a lower net load throughout the day and, consequently, attains the highest PR revenue. In contrast, although CS 4 records higher PR revenue than CSs 1 and 2 due to its higher net load, its participation in PR is limited to fewer time periods, leading to a relatively higher CA. Under scenarios of low net load, however, CS 4 demonstrates the potential to achieve both higher PR revenue and lower CA. Furthermore, CSs 2, 3, and 4 all realize PR revenue that exceeds their CA, thus enhancing overall profitability. Conversely, CS 1's PR revenue falls below its CA, resulting in a reduction in total profit. These results confirm that the CA method proposed herein can accurately quantify and distribute costs according to the number of DPR-participating units and the duration of their participation, thereby promoting a fair distribution of overall benefits among the CSs and IPPs. Moreover, incorporating CA into the CS dispatch process highlights the imperative for expediting flexibility retrofits of internal thermal power units to achieve a higher overall profit potential.

Table 4. Revenue and CA of PR auxiliary services on a typical winter day (10^4 CNY).

Power Source	CS1	CS2	CS3	CS4
Benefits	6.5678	8.8289	64.8350	11.2064
CA	8.4627	7.3259	12.0299	9.2492

To better compare the benefits brought by the three-stage optimization model, an independent optimization method is introduced as a benchmark. The differences between the two schemes are summarized in Table 5:

Table 5. Differences in optimization plans.

Scheme	CA	Objective Function Setting	Control Category
1	√	The Maximum Weighted Total Return of MCSs	Multi Coupling
2	×	The Maximum Benefit of a Single CS	Single Coupling

Scheme 1: Multi-stage economic dispatch optimization with CA for MCSs. The optimal power flow model of the main grid determines the active/reactive power dispatch demands at each time period, with the objective function maximizing the overall weighted revenue of the MCS. The optimization process considers CA and adjusted power output following the proposed operation model.

Scheme 2: Independent optimization for each CS without CA. The CA is calculated separately after sequentially optimizing each individual CS. In this scheme, the internal optimization of each CS aims to maximize its own intraday revenue based on known day-ahead active/reactive dispatch demands.

The optimal dispatch results for these two schemes are shown in Table 6:

Table 6. Comparison of CS benefits obtained from three-stage and independent optimizations (10^6 CNY).

Scheme	CS1	CS2	CS3	CS4
Typical summer day				
1	2.7173	2.0694	2.8643	5.3229
2	2.6938	2.0606	2.7716	5.1967
Typical winter day				
1	3.1085	1.7485	3.5280	5.1187
2	3.0953	1.7451	3.4671	5.0689

The three-stage optimization model yields higher profits for all CSs compared to independent optimization. The revenue increase is most significant for CS 3, with an improvement of over 3%. The revenue differences for CSs 1 and 2 are smaller compared to CSs 3 and 4. This discrepancy arises because CSs 3 and 4 are equipped with 600 MW units retrofitted with TPUMSTSR under RLVs, whereas CSs 1 and 2 only have 200 MW/300 MW conventional units, which rarely participate in DPR.

Under CA considerations, the output optimization potential of CSs 1 and 2 is limited. While cost differences exist, the overall profit impact remains minor due to limited revenue changes. In contrast, CSs 3 and 4, benefiting from lower CA through output adjustments, can significantly regulate unit output, thereby increasing DPR participation and related revenue opportunities. Notably, CS 4, with both 600 MW units having TPUMSTSR under RLVs, has the highest optimization potential, leading to the largest revenue increase.

Moreover, the aggregated revenue across all CSs indicates a significant difference in total economic benefits between summer and winter. This phenomenon highlights the impact of TPUMSTSR on optimization outcomes. DPR yields higher profits in winter due to a higher frequency of ramping opportunities and a larger revenue correction coefficient. Consequently, regardless of whether independent optimization or the proposed three-stage optimization is applied, thermal power units tend to adjust their output to maximize PR revenue, resulting in a relatively smaller optimization margin. Conversely, the influence of TPUMSTSR on the output range of thermal power units becomes more pronounced in summer when the revenue correction coefficient is lower. When incorporating CA in the three-stage optimization, TPUMSTSR more effectively modifies the output of thermal power units, thereby increasing PR revenue and expanding the optimization potential.

These findings demonstrate that, despite differences in the objective functions of the two optimization approaches, the proposed three-stage optimization model significantly enhances the economic efficiency of MCS operation, validating the effectiveness of the proposed dispatch strategy.

5. Conclusions

This study establishes an MCS operation model that incorporates TPUMSTSR under RLVs. The proposed method has been validated using operational data from the northeast China power grid. Based on the numerical results and corresponding analysis, the following conclusions are drawn:

- (1) The proposed TPUMSTSR model under RLVs combines the advantages of both RLVs and thermal storage retrofits. This model improves the real-time capability of thermal power units to compensate for the fluctuation of renewable generations, reduces the net load peak–valley difference, and enhances unit flexibility. As a result, the CS achieves higher overall operational revenue, improving economic efficiency. In the future, the operating costs of TPUMSTSR will be more accurately characterized to improve the accuracy of CS optimization methods.
- (2) The proposed three-stage economic dispatch optimization method independently optimizes the main grid and MCSs while preserving the privacy of each independent entity. By aligning with dispatch instructions from the main grid via optimal power flow, the method ensures coordinated operation. Furthermore, by introducing relaxation techniques, the approach simplifies the solution process for nonlinear objectives and constraints, thereby efficiently solving the mixed-integer nonlinear coordinated dispatch optimization problems. However, dividing the solution of nonlinear problems into multiple stages has a certain degree of simplification. We will study how to solve the optimization model more accurately through reinforcement learning algorithms.
- (3) The proposed dynamic CA method for PR revenue, which considers different time periods, enhances overall system profitability in scenarios where renewable energy penetration is high, coal-fired net load demand is low, and DPR compensation opportunities are frequent. By incorporating CA into the CS's dispatch process, this method provides a fair benefit distribution mechanism for MCSs while also encouraging thermal power unit flexibility retrofits to maximize overall system profitability. In the future, we will consider a wider range of application scenarios beyond MCSs and develop a more reasonable CA method.

Author Contributions: Conceptualization, N.Z. and Z.X.; methodology, Z.X., Y.C. and Y.Z.; software, Z.X. and X.D.; validation, Z.X.; formal analysis, Z.X.; investigation, Y.Z.; resources, Y.Z.; data curation, F.X.; writing—original draft preparation, Z.X.; writing—review and editing, N.Z. and Y.C.; visualization, Z.X.; supervision, Y.C.; project administration Y.C.; funding acquisition, Y.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research is supported by the National Key Research and Development Program of China (No. 2023YFB2405900).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Due to the nature of this research, participants of this study did not agree for their data to be shared publicly, so supporting data is not available. The data presented in this study are available on request from the corresponding author.

Acknowledgments: The authors thank Dongfeng Chang for assistance with the flexibility transformation of thermal power units.

Conflicts of Interest: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

1. Sonkar, P.; Rahi, O. Contribution of wind power plants in grid frequency regulation: Current perspective and future challenges. *Wind. Eng.* **2020**, *45*, 442–456.
2. Mogo, J.B.; Kamwa, I. Improved deterministic reserve allocation method for multi-area unit scheduling and dispatch under wind uncertainty. *J. Mod. Power Syst. Clean Energy* **2019**, *5*, 1142–1154.
3. Meegahapola, L.; Sguarezi, A.; Bryant, J.; Gu, M.; Conde, R.; Cunha, R. Power System Stability with Power-Electronic Converter Interfaced Renewable Power Generation: Present Issues and Future Trends. *Energies* **2020**, *13*, 3441.
4. Yang, L.J.; Zhou, N.C.; Zhou, G.P.; Chi, Y.; Wang, Q.G.; Lyu, X.M. An Accurate Ladder-Type Ramp Rate Constraint Derived From Field Test Data for Thermal Power Unit With Deep Peak Regulation. *IEEE Trans. Power Syst.* **2024**, *39*, 1408–1420.
5. Zou, Y.; Wang, Q.G.; Hu, B.; Chi, Y.; Zhou, G.P.; Xu, F.; Zhou, N.C.; Xia, Q.Q. Hierarchical evaluation framework for coupling effect enhancement of renewable energy and thermal power coupling generation system. *Int. J. Electr. Power Energy Syst.* **2023**, *146*, 108717.
6. Wang, Y.C.; Lou, S.H.; Wu, Y.W.; Wang, S.R. Flexible operation of retrofitted thermal power units to reduce wind curtailment considering thermal energy storage. *IEEE Trans. Power Syst.* **2020**, *35*, 1178–1187.
7. Yang, L.J.; Zhou, N.C.; Hu, B.; Chen, L.; Xu, F.; Wang, Q.G.; Qu, Z. Optimal scheduling method for coupled system based on ladder-type ramp rate of thermal power units. *Proc. CSEE* **2022**, *42*, 153–164.
8. Lee, S.; Kim, J.; Tahmasebi, A.; Jeon, C.; Liu, Y.; Yu, J. Comprehensive technical review of the high-efficiency low-emission technology in advanced thermal power units. *Rev. Chem. Eng.* **2021**, *39*, 363–386.
9. Luo, G.L.; Zhang, X.; Liu, S.S.; Dan, E.L.; Guo, Y.W. Demand for flexibility improvement of thermal power units and accommodation of wind power under the situation of high-proportion renewable integration—taking north Hebei as an example. *Environ. Sci. Pollut. Res.* **2019**, *26*, 7033–7047.
10. Wang, C.Y.; Song, J.W. Performance assessment of the novel coal-fired combined heat and power plant integrating with flexibility renovations. *Energy* **2023**, *263*, 125886.
11. Liu, M.; Ma, G.F.; Wang, S.; Wang, Y.; Yan, J.J. Thermo-economic comparison of heat–power decoupling technologies for combined heat and power plants when participating in a power-balancing service in an energy hub. *Renew. Sustain. Energy Rev.* **2021**, *152*, 111715.
12. Blanquiceth, J.; Carademil, J.M.; Henríquez, M.; Escobar, R. Thermodynamic evaluation of a pumped thermal electricity storage system integrated with large-scale thermal power plants. *Renew. Sustain. Energy Rev.* **2023**, *175*, 113134.
13. Liu, Z.F.; Wang, C.Y.; Fan, J.L.; Liu, M.; Xing, Y.; Yan, J.J. Enhancing the flexibility and stability of coal-fired power plants by optimizing control schemes of throttling high-pressure extraction steam. *Energy* **2024**, *288*, 129756.
14. Cui, R.Y.; Hultman, N.; Cui, D.Y.; Mcjeon, H.; Yu, S.; Edwards, M.R. A plant-by-plant strategy for high-ambition coal power phaseout in China. *Nat. Commun.* **2021**, *12*, 1468.
15. Hentschel, J.; Zindler, H.; Spliethoff, H. Modelling and transient simulation of a supercritical coal-fired power plant: Dynamic response to extended secondary control power output. *Energy* **2017**, *137*, 927–940.
16. Garbrecht, O.; Bieber, M.; Kneer, R. Increasing fossil power plant flexibility by integrating molten-salt thermal storage. *Energy* **2017**, *118*, 876–883.
17. Zhao, Y.L.; Wang, C.Y.; Liu, M.; Chong, D.T.; Yan, J.J. Improving operational flexibility by regulating extraction steam of high-pressure heaters on a 660 MW supercritical coal-fired power plant: A dynamic simulation. *Appl. Energy* **2018**, *212*, 1295–1309.
18. Zima, W.; Gradziel, S.; Cebula, A.; Rerak, M.; Kozak-Jagiela, E.; Pilarczyk, M. Mathematical model of a power boiler operation under rapid thermal load changes. *Energy* **2023**, *263*, 125836.
19. Zhao, Y.; Liu, M.; Wang, C.; Li, X.; Chong, D.; Yan, J. Increasing operational flexibility of supercritical coal-fired power plants by regulating thermal system configuration during transient processes. *Appl. Energy* **2018**, *228*, 2375–2386.
20. Gulotta, F.; Daccò, E.; Bosisio, A.; Falabretti, D. Opening of Ancillary Service Markets to Distributed Energy Resources: A Review. *Energies* **2023**, *16*, 2814.
21. Heendeniya, C.; Sumper, A.; Eicker, U. The multi-energy system co-planning of nearly zero-energy districts—Status-quo and future research potential. *Appl. Energy* **2020**, *267*, 114953.
22. Yang, Y.D.; Gao, H.J.; Xiang, Y.M.; Guo, M.H.; Wang, J.Y.; Liu, J.Y. Coordinated Generation Scheduling Considering Peak Regulation Cost Allocation and Compensation Distribution in Power Systems. *J. Mod. Power Syst. Clean Energy* **2025**, *20*, 1–12.
23. Yang, B.; Cao, X.Y.; Cai, Z.H.; Yang, T.G.; Chen, D.W.; Gao, X.H. Unit Commitment Comprehensive Optimal Model Considering the Cost of Wind Power Curtailment and Deep Peak Regulation of Thermal Unit. *IEEE Access* **2020**, *8*, 71318–71325.

24. Bavafa, F.; Niknam, T.; Azizipanah-Abarghooee, R.; Terzija, V. A New Bi-Objective Probabilistic Risk Based Wind-Thermal Unit Commitment Using Heuristic Techniques. *IEEE Trans. Ind. Inform.* **2017**, *13*, 115–124.
25. Zhang, Y.; Yao, F.; Herbert, H.C.; Fernando, T.; Trinh, H. Wind-thermal systems operation optimization considering emission problem. *Int. J. Electr. Power Energy Syst.* **2015**, *65*, 238–245.
26. Ye, Z.; Li, X.Q.; Jiang, F.; Chen, L.; Wang, Y.L.; Dai, S.F. Hierarchical optimization economic dispatching of combined wind-PV-thermal-energy storage system considering the optimal energy abandonment rate. *Power Syst. Technol.* **2021**, *45*, 2270–2280.
27. Wu, H.Y.; Shahidehpour, M.; Alabdulwahab, A.; Abusorrah, A. Thermal Generation Flexibility with Ramping Costs and Hourly Demand Response in Stochastic Security-Constrained Scheduling of Variable Energy Sources. *IEEE Trans. Power Syst.* **2015**, *30*, 2955–2964.
28. Zhang, R.F.; Yan, K.F.; Li, G.Q.; Jiang, T.; Li, X.; Chen, H.H. Privacy-preserving decentralized power system economic dispatch considering carbon capture power plants and carbon emission trading scheme via over-relaxed ADMM. *Int. J. Electr. Power Energy Syst.* **2020**, *121*, 106094.
29. Reddy, S.S. Optimal scheduling of thermal-wind-solar power system with storage. *Renew. Energy* **2017**, *101*, 1357–1368.
30. Energiewende, A. Flexibility in Thermal Power Plants-with a Focus on Existing Coal-Fired Power Plants; Agora Energiewende: Berlin, Germany, 2017.
31. Xu, T.; Wang, H. Impact of flexible operation on the lifetime of thermal power plant components. *Energy* **2020**, *192*, 116634.
32. Benato, A.; Bracco, S.; Stoppato, A.; Mirandola, A. LTE: A procedure to predict power plants dynamic behaviour and components lifetime reduction during transient operation. *Appl. Energy* **2016**, *162*, 880–891.
33. Wang, C.Y.; Liu, M.; Li, B.X.; Liu, Y.W.; Yan, J.J. Thermodynamic analysis on the transient cycling of coal-fired power plants: Simulation study of a 660 MW supercritical unit. *Energy* **2017**, *122*, 505–527.
34. Wang, C.; Zhang, X. Impact of load-following operation on coal consumption in thermal power plants. *Energy Convers. Manag.* **2018**, *165*, 1–10.
35. Peng, Y.; Lou, S.; Fan, Y.; Wu, Y.W.; Liang, S.H. Low-carbon Economical Dispatch of Power System Considering Thermal Energy Storage in Thermal Power Units. *Power Syst. Technol.* **2020**, *44*, 3339–3345.
36. Wamalwa, F.; Ishimwe, A. Optimal energy management in a grid-tied solar PV-battery microgrid for a public building under demand response. *Energy Rep.* **2024**, *12*, 3718–3731.
37. Ma, H.Y.; Yan, Z.; Li, M.J.; Han, D.; Han, X.; Song, Y.Q. Benefit evaluation of the deep peak-regulation market in the northeast China grid. *CSEE J. Power Energy Syst.* **2019**, *5*, 533–544.
38. Zhou, Y.Y.; Wang, Q.G.; Zou, Y.; Chi, Y.; Zhou, N.C.; Zhang, X.F. Voltage Profile Optimization of Active Distribution Networks Considering Dispatchable Capacity of 5G Base Station Backup Batteries. *J. Mod. Power Syst. Clean Energy* **2023**, *11*, 1842–1856.
39. Li, J.H.; Zhang, J.H.; Mu, G.; Yan, G.G.; Shi, S.J. Hierarchical Optimization Scheduling of Deep Peak Shaving for Energy-Storage Auxiliary Thermal Power Generating Units. *Power Syst. Technol.* **2019**, *43*, 3961–3970.

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.